

## ***Direct Memory Access Controller (DMA) Module***

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This chapter describes the direct memory access (DMA) controller.

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## 16.1 Overview

The DMA controller is used to transfer data between two locations in the memory map in the background of CPU operations. Typically, the DMA is used to:

- Transfer blocks of data between external and internal data memories
- Restructure portions of internal data memory
- Continually service a peripheral
- Page program sections to internal program memory

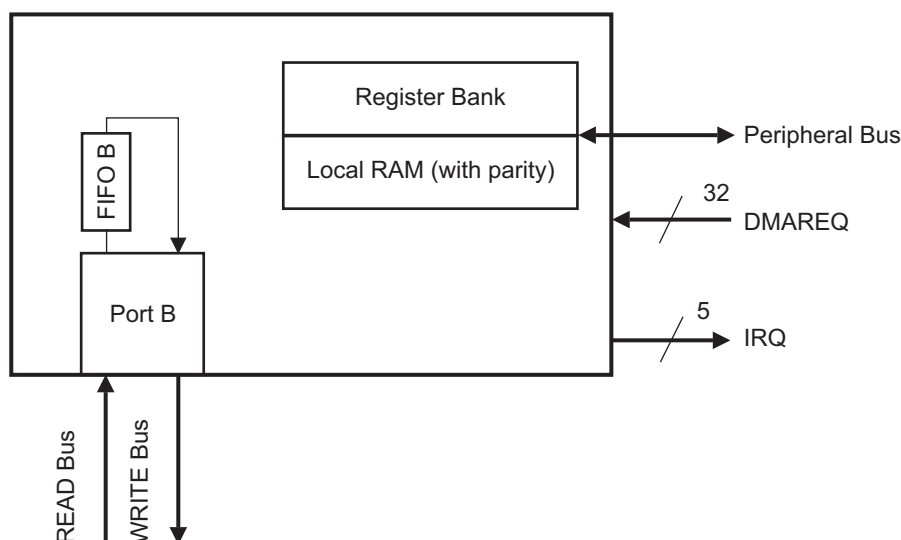
### 16.1.1 Main Features

- CPU independent data transfer
- One master port - PortB (64-bits wide) that interfaces microcontrollers Memory System.
- FIFO buffer (4 entries deep and each 64-bits wide)
- Channel control information is stored in RAM protected by parity
- 16 channels with individual enable
- Channel chaining capability
- 32 peripheral DMA requests
- Hardware and Software DMA requests
- 8-, 16-, 32-, or 64-bit transactions supported
- Multiple addressing modes for source/destination (fixed, increment, offset)
- Auto-initiation
- Power-management mode
- Memory Protection for the address range DMA can access with four configurable memory regions

#### 16.1.1.1 Block Diagram

Figure 16-1 gives a detailed view of the DMA internal architecture. DMA data read and write access happens through Port B. FIFO B is 4 levels deep and 64-bits wide. 32 DMA requests go into the DMA that can trigger DMA transfers. Five interrupt request lines go out of the DMA to signal that a certain transfer status is reached. Register banks hold the memory-mapped DMA configuration registers. Local RAM consists of DMA control packets and is secured by parity. All the programming / configuration of the DMA controller is done via the Peripheral bus.

**Figure 16-1. DMA Block Diagram**



## 16.2 Module Operation

The DMA acts as an independent master in the platform architecture. All DMA memory and register accesses are performed in user mode. If the DMA writes to registers that are only accessible in privileged mode, the write will not be performed.

The DMA registers and its local RAM can only be accessed in privilege mode. Therefore, it is not possible for the DMA to reprogram itself.

### 16.2.1 Memory Space

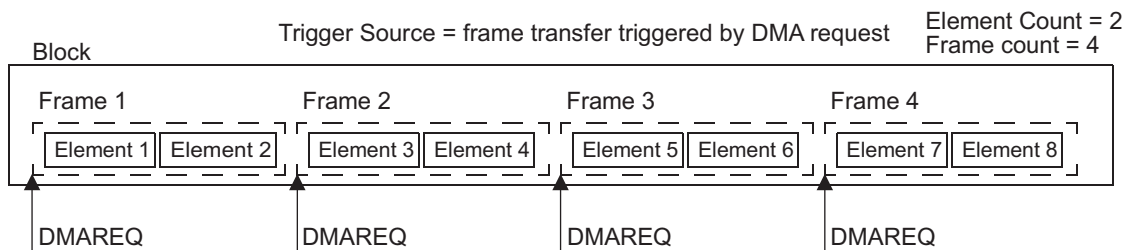
The DMA controller makes no distinction between program memory and data memory. The DMA controller can transfer to and from any space within the 4 gigabyte physical address map, by programming the absolute address for the source and destination in the control packet. Control packets store the transfer information such as source address, destination address, transfer count and control attributes for each channel.

### 16.2.2 DMA Data Access

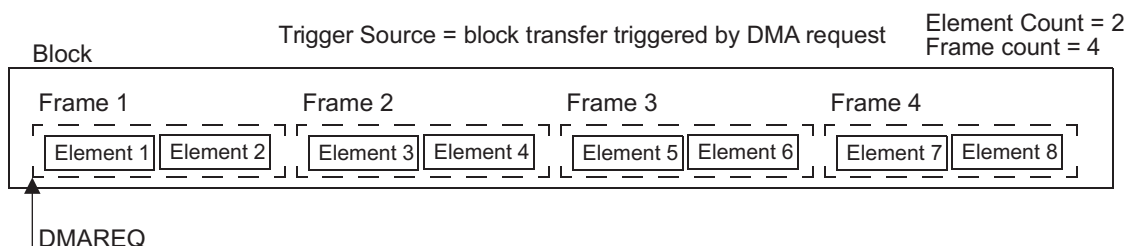
The DMA controller refers to data in three levels of granularity:

- **Element:** Depending on the programmed data type, an 8-bit, 16-bit, 32-bit, or a 64-bit value. The type can be individually selected for the source (read) and destination (write). See [Figure 16-2](#) and [Figure 16-3](#) for an example of the use of elements. An element transfer cannot be interrupted.
- **Frame:** One or more elements to be transferred as a unit. A frame transfer can be interrupted between element transfers. See [Figure 16-2](#) for an example. Use a frame size of one and frame transfer trigger source for transfers of one element per request.
- **Block:** One or more frames to be transferred as a unit. Each channel can transfer one block of data (once or multiple times). See [Figure 16-3](#) for an example.

**Figure 16-2. Example of a DMA Transfer Using Frame Trigger Source**



**Figure 16-3. Example of a DMA Transfer Using Block Trigger Source**



### 16.2.3 Addressing Modes

There are three addressing modes supported by the DMA controller that can be setup independent for the source and the destination address:

- Constant -- source and/or destination addresses do not change.
- Post incremented -- source and/or destination address are post-incremented by the element size.
- Indexed -- source and/or destination address is post-incremented as defined in the Element Index Offset Register ([Section 16.3.2.5](#)) and the Frame Index Offset Register ([Section 16.3.2.6](#)).

An unaligned address with respect to the element size is not supported.

### 16.2.4 DMA Channel Control Packets

There are a total of 16 control packets. Each control packet is associated with a channel in a fixed order. For example, control packet 0 stores channel information for channel 0. The DMA requests can be mapped to the individual channels as described in [Section 16.2.7](#). The mapping scheme between DMA requests and channels is shown in [Figure 16-4](#). Each control packet contains nine fields. The first six fields compose the primary control packet and are programmable during DMA setup. The last three fields compose working control packet and are only readable by the CPU. The working control packets are used to support auto-initiation. The organization of control packets is shown in [Figure 16-5](#).

The primary control packet contains channel information such as source address, destination address, transfer count, element/frame offset value and channel configuration. Source address, destination address and transfer count also have their respective working images. The three fields of working images compose a working control packet and are not accessible to the CPU in write access.

The first time a DMA channel is selected for a transaction, the following process occurs:

1. The primary control packet is first read by the DMA state machine.
2. Once the channel is arbitrated, the current source address, destination address and transfer count are then copied to their respective working images.
3. When the channel is serviced again by the DMA, the state machine will read both the primary control packet and the working control packet to continue the DMA transaction until the end of an entire block transfer.

When the same channel is requested again, the state machine will start again by reading only the primary control packet and then continue the same process described above. The user software need not set up control packets again because the contents of the primary control packet were never lost. The working images of the control packets are reducing the software overhead and interaction with the DMA module to a minimum.

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**NOTE:** Changing the contents of a channel control packet will clear the corresponding pending bit ([Section 16.3.1.2](#)) if the channel has a pending status. If the control packet of an active channel (as indicated in [Section 16.3.1.3](#)) is changed, then the channel will stop immediately at an arbitration boundary. When the same channel is triggered again, it will begin with the new control packet information.

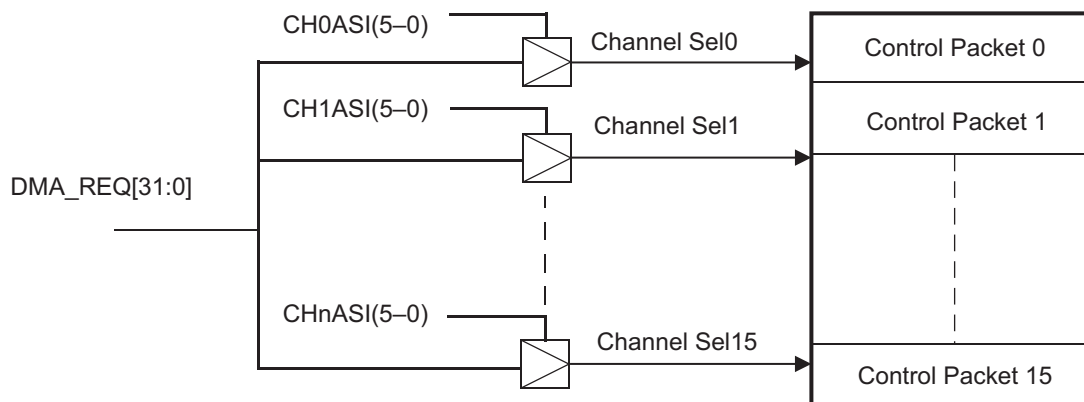
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#### 16.2.4.1 Initial Source Address

This field stores the absolute 32-bit source address of the DMA transfer.

#### 16.2.4.2 Initial Destination Address

This field stores the absolute 32-bit destination address of the DMA transfer.

**Figure 16-4. DMA Request Mapping and Control Packet Organization**

**Figure 16-5. Control Packet Organization and Memory Map**

|             | Base + 0XXX0           | Base + 0XXX4                | Base + 0XXX8           |               | Base + 0XXXXC |
|-------------|------------------------|-----------------------------|------------------------|---------------|---------------|
| Base + 0x00 | Initial Source Address | Initial Destination Address | Initial Transfer Count | } Primary CP0 | Reserved      |
| 0x10        | Channel Configuration  | Element Offset Value        | Frame Offset Value     |               |               |
| 0x20        | Initial Source Address | Initial Destination Address | Initial Transfer Count | } Primary CP1 |               |
| 0x30        | Channel Configuration  | Element Offset Value        | Frame Offset Value     |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
| 0x1E0       | Initial Source Address | Initial Destination Address | Initial Transfer Count | } Primary CPn |               |
| 0x1F0       | Channel Configuration  | Element Offset Value        | Frame Offset Value     |               |               |
|             | Reserved               | Reserved                    | Reserved               |               |               |
| 0x800       | Current Source Address | Current Destination Address | Current Transfer Count | } Working CP0 |               |
| 0x810       | Current Source Address | Current Destination Address | Current Transfer Count |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
|             | ⋮                      | ⋮                           | ⋮                      |               |               |
| 0x8F0       | Current Source Address | Current Destination Address | Current Transfer Count | } Working CPn |               |

#### 16.2.4.3 Initial Transfer Count

The transfer count field is composed of two parts. The frame transfer count value and the element transfer count value. Each count value is 13 bits wide. As a Single Block transfer maximum of 512 Mbytes of data can be transferred. Element count and frame count are programmed according to the source data structure.

The total transfer size is calculated as:

$$T_{sz} = E_{rsz} \cdot E_{tc} \cdot F_{tc} \quad (26)$$

where

$T_{sz}$  = Total Transfer Size

$E_{rsz}$  = Read Element Size

$E_{tc}$  = Element Transfer Count

$F_{tc}$  = Frame Transfer Count

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**NOTE:** A zero element count with a non-zero frame count or a non-zero element count with a zero frame count are all considered as zero total transfer count. No DMA transaction is initiated with any of the counters set to 0.

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#### 16.2.4.4 Channel Configuration Word

The channel configuration defines the following individual parameters

- Read element size
- Write element size
- Trigger type (frame or block)
- Addressing mode for source
- Addressing mode for destination
- Auto-initiation mode
- Next control packet to be triggered at control packet finish (Channel Chaining)

#### 16.2.4.5 Element/Frame Offset Value

There are 4 offset values that allow the creation of different types of buffers in RAM and address registers in a structured manner: an element offset value for source and destination and a frame offset value for source and destination.

The element offset value for source and/or destination defines the offset to be added after each element transfer to the source and/or destination address. The frame offset value for source and/or destination defines the offset to be added to the source and/or destination address after the element count reaches zero. The element and frame offset values must be defined in terms of the number of bytes of offset. The DMA controller does not adjust the element/frame index number according to the element size. An index of 2 means *increment the address by 2* and not by 16 when the element size is 64 bits.

#### 16.2.4.6 Current Source Address

The current source address field contains the current working source address during a DMA transaction. The current source address is incremented during post increment addressing mode or indexing mode.

#### 16.2.4.7 Current Destination Address

The current destination address field contains the current working destination address during a DMA transaction. The current destination address is incremented during post-increment addressing mode or indexing mode.

### 16.2.4.8 Current Transfer Count

The current transfer count stores the remaining number of elements to be transferred in a block. It is decremented by one for each element read from the source location.

Figure 16-6, Figure 16-7, and Figure 16-8 show some examples of DMA transfers.

**Figure 16-6. DMA Transfer Example 1**

| Source |    |    |    | Destination |     |          |          |
|--------|----|----|----|-------------|-----|----------|----------|
|        | f1 | f2 | f3 | f4          |     |          |          |
| 0x00   | E1 | E3 | E5 | E7          | 0x0 | E1/3/5/7 | E2/4/6/8 |
| 0x04   |    |    |    |             | 0x4 |          |          |
| 0x08   |    |    |    |             |     |          |          |
| 0x0C   | E2 | E4 | E6 | E8          | 0x0 | E1/2     | E3/4     |
|        |    |    |    |             |     | E5/6     | E7/8     |
|        |    |    |    |             |     |          |          |
|        |    |    |    |             | 0x0 | E1       | E3       |
|        |    |    |    |             | 0x4 | E2       | E4       |
|        |    |    |    |             |     | E5       | E7       |
|        |    |    |    |             |     | E6       | E8       |
|        |    |    |    |             |     |          |          |
|        |    |    |    |             | 0x0 | E1       | E2       |
|        |    |    |    |             | 0x4 | E5       | E6       |
|        |    |    |    |             |     | E7       | E8       |
|        |    |    |    |             |     |          |          |

Source Element Index = 12

Source Frame Index = 1

Dest. Element Index = 1

Dest. Frame Index = 0

Dest. Element Index = 0

Dest. Frame Index = 1

Dest. Element Index = 4

Dest. Frame Index = 1

Dest. Element Index = 1

Dest. Frame Index = 2

The example assumes the following setup.

Read Element Size = 8 bit

Write Element Size = 8 bit

Element Count = 2

Frame Count = 4

**Figure 16-7. DMA Indexing Example 1**

|      | f1 | f2 | f3  | f4  |
|------|----|----|-----|-----|
| 0x0  | E1 | E5 | E9  | E13 |
| 0x10 | E2 | E6 | E10 | E14 |
| 0x20 | E3 | E7 | E11 | E15 |
| 0x30 | E4 | E8 | E12 | E16 |

Element Index = 16

Frame Index = 4

This example can be applied to either source or destination indexing and assumes the following setup.

Element Size = 16 bit

Element Count = 4

Frame Count = 4

**Figure 16-8. DMA Indexing Example 2**

|      |    |    |    |     |     |     |     |     |
|------|----|----|----|-----|-----|-----|-----|-----|
| 0x0  | E1 | E4 | E7 | E10 | E13 | E16 | E19 | E22 |
| 0x20 |    |    |    |     |     |     |     |     |
| 0x40 | E2 | E5 | E8 | E11 | E14 | E17 | E20 | E23 |
| 0x60 |    |    |    |     |     |     |     |     |
| 0x80 | E3 | E6 | E9 | E12 | E15 | E18 | E21 | E23 |

Element Index = 64

Frame Index = 4

This example can be applied to either source or destination indexing and assumes the following setup.

Element Size = 32 bit

Element Count = 3

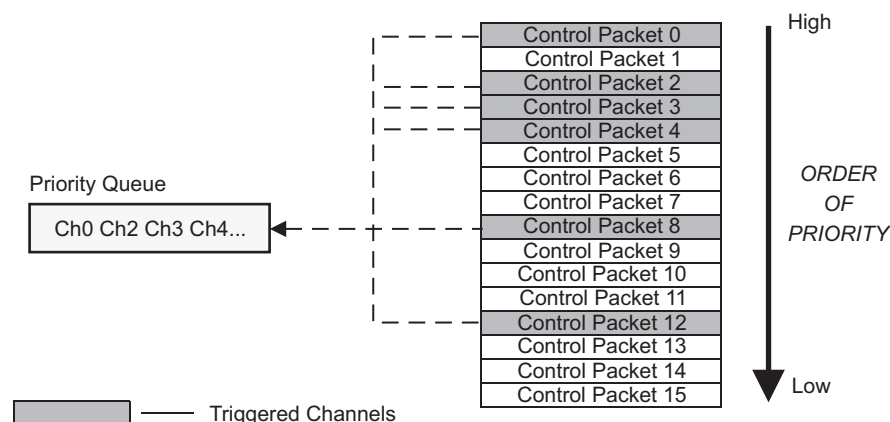
Frame Count = 8

### 16.2.5 Priority Queue

User can assign channels in to priority queues to facilitate request handling during arbitration. The port has two priority queues: a high and a low priority queue. The queue can be configured to follow a fixed or rotating priority scheme. Fixed priority is such that the lower the channel number (Figure 16-9), the higher its priority. Rotating priority is based on a round-robin scheme. Initially, the priority list is sorted according to the fixed priority scheme. Channels assigned to the high priority queue are always serviced first according to the selected priority scheme before channels in the low priority queue are serviced. Table 16-1 describes how arbitration is performed according to different priority schemes.

**NOTE:** Since the DMA controller provides the capability to map any one of the 32 hardware DMA request lines to any channel, the numerical order of the hardware DMA request does not imply any priority. The priority of each hardware DMA request is programmed and determined by software.

**Figure 16-9. Fixed Priority Scheme**

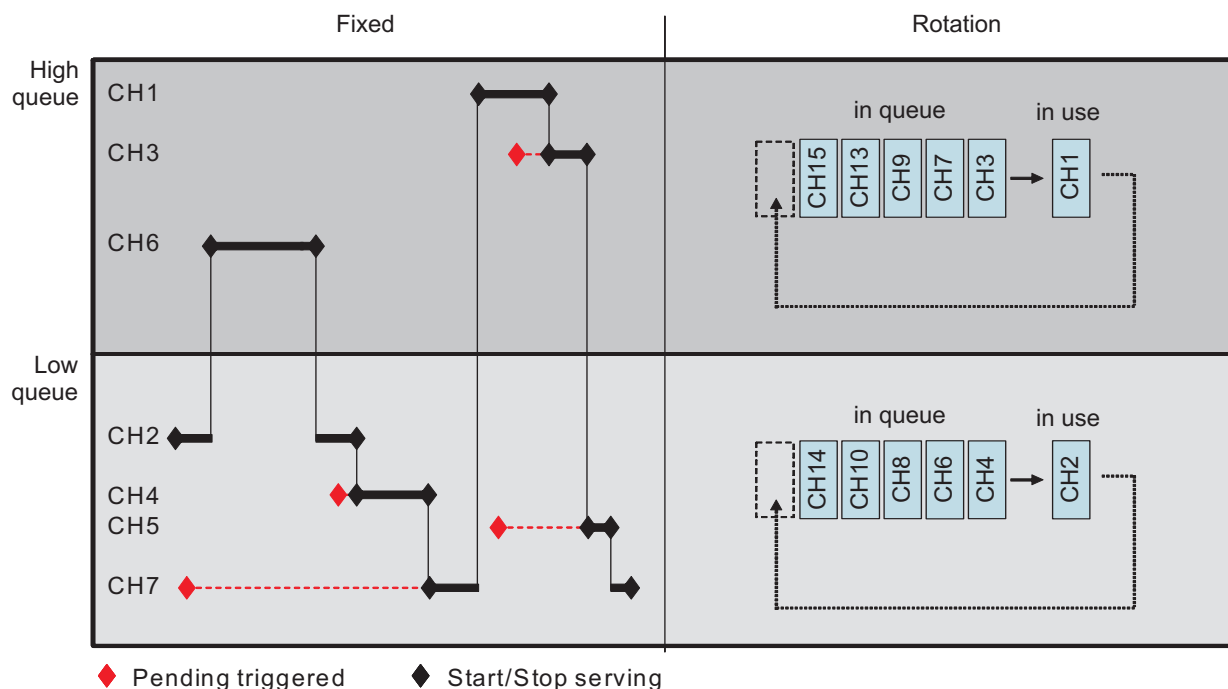


The above figure illustrates that by default Lower the channel number, higher the Priority.

**Table 16-1. Arbitration According to Priority Queues and Priority Schemes**

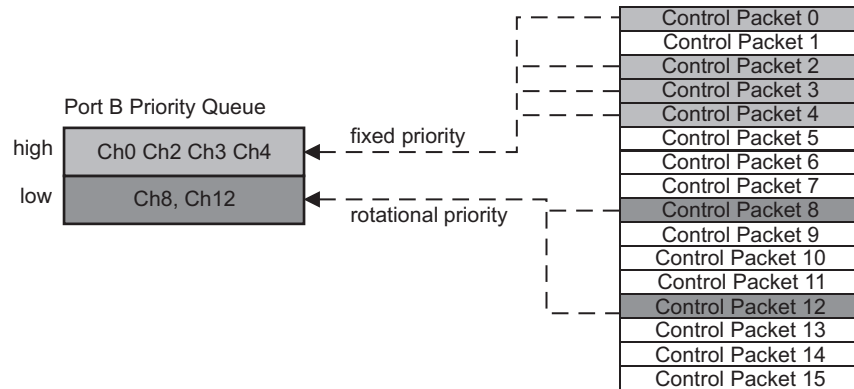
| Queue         | Priority Scheme | Remark                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High priority | Fixed           | Channels are serviced in an ascending order according to the channel number. The lower the channel number, the higher the priority. A channel will be arbitrated out whenever there is a higher pending channel. Otherwise a channel is completely serviced until its transfer count reaches zero before the next highest pending channel is serviced. When there is no pending channels left in high queue then the DMA switches to service low queue channels.                                                                                           |
|               | Rotating        | Channels are arbitrated by using the round-robin scheme. Arbitration is performed when the FIFO is empty. When there are no pending channels left in high queue then the DMA switches to service low queue channels.                                                                                                                                                                                                                                                                                                                                       |
| Low priority  | Fixed           | Channels are serviced in an ascending order according to the channel number. The lower the channel number the higher the priority. A channel will be arbitrated out whenever there is a higher-priority pending channel. Otherwise a channel is completely serviced until its transfer count reaches zero, before the next highest pending channel is serviced. If there is a pending channel in the high-priority queue while DMA is servicing a low queue channel then DMA will switch back to service high queue channel after an arbitration boundary. |
|               | Rotating        | Channels are arbitrated by using round-robin scheme. Arbitration is performed when the FIFO is empty.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

A Simple Priority Queues example in both Fixed and Rotation Scheme is shown in [Figure 16-10](#).

**Figure 16-10. Example of Priority Queues**


For optimal system performance, the high priority channels should be put in fixed arbitration scheme and low priority channels in the rotating priority scheme as illustrated in [Figure 16-11](#).

**Figure 16-11. Example Channel Assignments**



1 The above figure illustrates the channel assignments in a system with 16 channels. This approach can be scaled dependent on the total channels available.

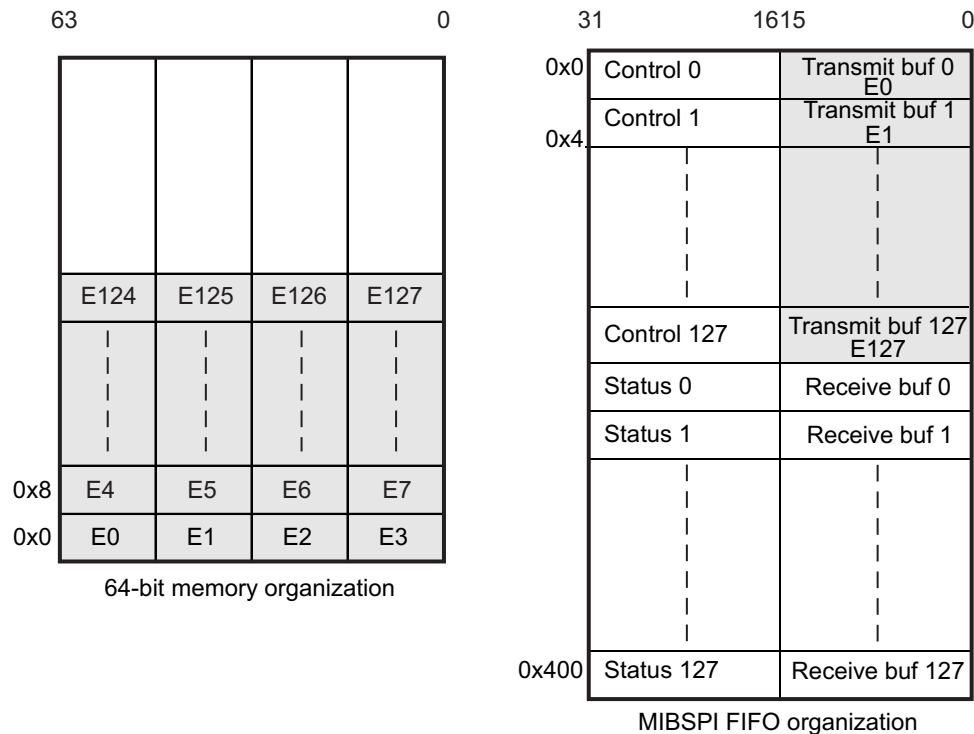
### 16.2.6 Data Packing and Unpacking

The DMA controller automatically performs the necessary data packing and unpacking when the read element size differs from the write element size. Data packing is required when the read element size is smaller than the write element size; data unpacking is required when the read element size is larger than the write element size. When the read element size is equal to the write element size, no packing is performed during read, nor is any unpacking performed during write.

[Figure 16-12](#) shows an example of data unpacking in which the DMA is used to transfer 128 transmit data elements to the MibSPI FIFO buffer. In this example, data unpacking is required because the read element size is 64 while the write element size is 16. The DMA first performs a 64-bit read from the source into its FIFO buffer. After the 64-bit data is read into the DMA FIFO buffer, it must unpack the data into four 16-bit data elements before writing out to the destination. Therefore the DMA would need to perform four 16-bit write operations to the destination.

**NOTE:** In the example in [Figure 16-12](#), to transmit data at the lower bits of the MibSPI, bits 15:0, the destination address should be incremented by a factor of 2.

- NOTE:**
- 1) The element Count ([Section 16.3.2.3](#)) refers only to the read element.
  - 2) Data unpacking does not require the DMA request. Once the DMA request is received, data from Source is moved in to FIFO and unpacking happens until the FIFO is empty.
  - 3) DMA assumes the destination is always ready and will perform write immediately. In case of data unpacking and Constant Addressing Mode write ([Section 16.3.2.4](#) (1 - 0) = 0) the destination data will be overwritten by next data or next data might be skipped in case the destination has overflow protection (eg., SCITD register). User should configure DMA to avoid data unpacking if the Destination is configured as Constant Addressing Mode write to avoid data loss.

**Figure 16-12. Example of DMA Data Unpacking**


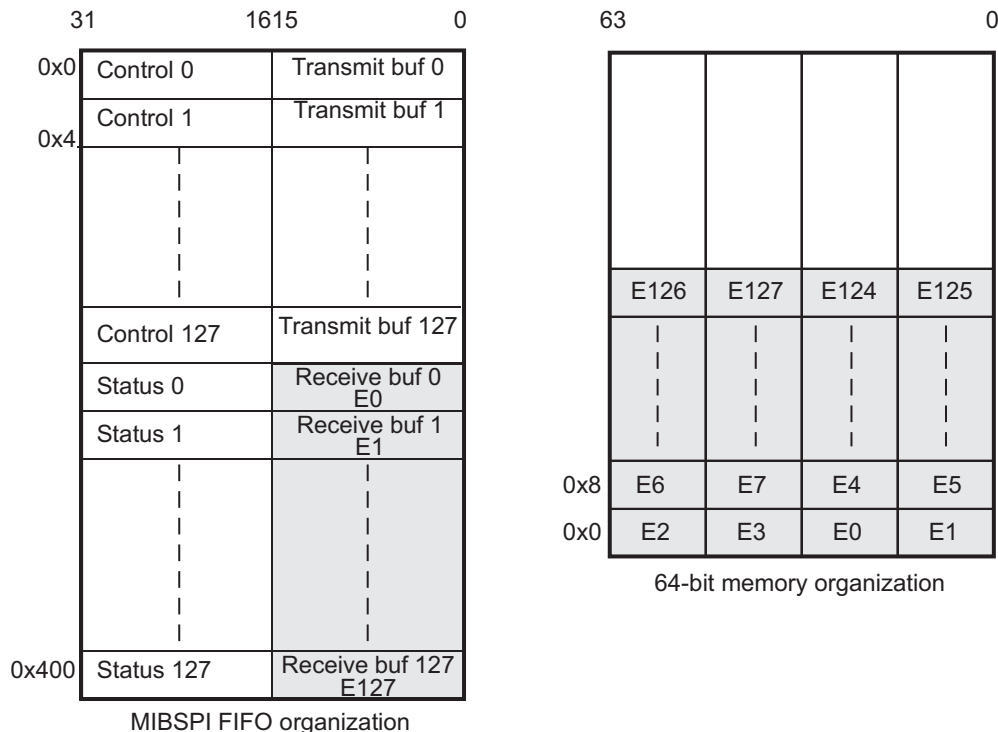
In this example, initialization of the MIBSPI FIFO is illustrated and assumes the following setup:

Read Element Size = 64 bit  
Write Element Size = 16 bit  
Element Count = 32  
Frame Count = 1  
Source Element Index = n/a, use post increment addressing mode  
Source Frame Index = n/a, use post increment addressing mode  
Destination Element Index = 4  
Destination Frame Index = 0

When the read element size is smaller than the write element size, the DMA controller needs to perform data packing. The number of elements to pack is equal to the ratio between the write element size and read element size. In the example in [Figure 16-13](#), the read element size is 16 bits and the write element size is 64 bits. The DMA controller would first pack the first four elements by performing four consecutive 16-bit read accesses of E0, E1, E2, and E3 into the first word of the DMA's internal FIFO. The DMA controller would then perform one single 64-bit write operation to transfer the data to the 64-bit destination memory.

Normally, the DMA controller carries out bus transactions on the bus according to the element size. For example, the DMA controller would perform a 16-bit read transaction if the read element size is programmed as 16 bits, or an 8-bit write transaction if the write element size is programmed as 8 bit. The exception is when the total transfer size is as defined in [Equation 26](#) is not a multiple of the write element size.

**Figure 16-13. Example of DMA Data Packing**



In this example, a read of the MIBSPI FIFO is illustrated and assumes the following setup:

Read Element Size = 16 bit  
Write Element Size = 64 bit  
Element Count = 128  
Frame Count = 1  
Source Element Index = 4  
Source Frame Index = 0  
Destination Element Index = n/a, use post increment addressing mode  
Destination Frame Index = n/a, use post increment addressing mode

For example, if the read element size is 8 bits, the element transfer count is equal to 9, and the write element size is 64 bit. The DMA controller would first perform eight 8-bit read transactions from the source. It would then perform a 64-bit write to the destination. When the same channel wins arbitration again, the DMA controller would first perform one 8-bit read from the source, followed by one 8-bit write to the destination, even though the write element size is 64 bit.

**NOTE:** Since peripherals are slower, it is advised to use data packing feature with caution for reading data from peripherals. Improper use might delay servicing other pending DMA channels.

## 16.2.7 DMA Request

There are three ways to start a DMA transfer:

- **Software request:** The transfer will be triggered by writing to SW Channel Enable Set and Status Register (Section 16.3.1.6). The software request can trigger either a block or a frame transfer depending on the setting of the TTYPE bit in the Channel Control Register (Section 16.3.2.4).
- **Hardware request:** The DMA controller can handle up to 32 DMA Request lines. A hardware request can trigger either a frame or a block transfer depending on the setting of the TTYPE bit in the Channel Control Register (Section 16.3.2.4).
- **Triggered by other control packet:** When a control packet finishes the programmed number of transfers it can trigger another channel to initiate its transfers.

Each time a DMA request is made, either one frame transfer or one block transfer can be chosen. An active DMA request signal will trigger a DMA transaction.

The DMA controller has a two-level buffer to capture HW requests per channel. When a HW request is generated and the channel is enabled, the corresponding bit in the DMA Status Register (Section 16.3.1.3) is set. The pending register acts as a first-level buffer. Typically, a peripheral acting as a source of a transfer would initiate another request after its data registers have been read out by DMA, even though that data has not been completely transferred to the destination. If a second HW request is generated by the peripheral, the DMA controller has an extra request buffer to capture this second request and service it after the first request is complete.

---

**NOTE:** The DMA cannot capture more than three requests if its request buffers are already full. If any request occur during this moment DMA will discard it.

---

The DMA controller also supports a mix of hardware and software requests on the same channel. Note that such interchangeable usage may result into an out of sync for DMA channel and peripheral. The application needs to be careful as the DMA does not have a built-in mechanism to protect against this loss of synchronization.

If a software request is generated, the corresponding bit in the Channel Pending Register (Section 16.3.1.2) is set accordingly. If the pending request is not completely serviced by the DMA and a hardware request is generated by a peripheral onto the same channel, the DMA will capture and recognize this hardware request into its request buffer.

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**NOTE:** The DMA controller cannot recognize two software requests on the same channel if the first software request is still pending. If such request occur DMA will discard it. Therefore the user software should check the pending register before issuing a new software request.

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The DMA module has 16 channels and up to 32 hardware DMA requests. The module contains DREQS<sub>ix</sub> registers that are used to map the DMA requests to the DMA channels. By default, channel 0 is mapped to request 0, channel 1 to request 1, and so on.

Some DMA requests have multiple sources, as shown in Table 16-2. The application must ensure that only one of these DMA request sources is enabled at any time.

**Table 16-2. DMA Request Line Connection**

| Modules                   | DMA Request Sources                 | DMA Request |
|---------------------------|-------------------------------------|-------------|
| MIBSPI1                   | MIBSPI1[1] <sup>(1)</sup>           | DMAREQ[0]   |
| MIBSPI1                   | MIBSPI1[0] <sup>(2)</sup>           | DMAREQ[1]   |
| SPI2                      | SPI2 receive                        | DMAREQ[2]   |
| SPI2                      | SPI2 transmit                       | DMAREQ[3]   |
| MIBSPI1 / MIBSPI3 / DCAN2 | MIBSPI1[2] / MIBSPI3[2] / DCAN2 IF3 | DMAREQ[4]   |
| MIBSPI1 / MIBSPI3 / DCAN2 | MIBSPI1[3] / MIBSPI3[3] / DCAN2 IF2 | DMAREQ[5]   |
| DCAN1 / MIBSPI5           | DCAN1 IF2 / MIBSPI5[2]              | DMAREQ[6]   |

<sup>(1)</sup> SPI1, SPI3, SPI5 receive in standard SPI mode

<sup>(2)</sup> SPI1, SPI3, SPI5 transmit in standard SPI mode

**Table 16-2. DMA Request Line Connection (continued)**

| Modules                                  | DMA Request Sources                                                              | DMA Request |
|------------------------------------------|----------------------------------------------------------------------------------|-------------|
| MIBADC1 / MIBSPI5                        | MIBADC1 event / MIBSPI5[3]                                                       | DMAREQ[7]   |
| MIBSPI1 / MIBSPI3 / DCAN1                | MIBSPI1[4] / MIBSPI3[4] / DCAN1 IF1                                              | DMAREQ[8]   |
| MIBSPI1 / MIBSPI3 / DCAN2                | MIBSPI1[5] / MIBSPI3[5] / DCAN2 IF1                                              | DMAREQ[9]   |
| MIBADC1 / I2C / MIBSPI5                  | MIBADC1 G1 / I2C receive / MIBSPI5[4]                                            | DMAREQ[10]  |
| MIBADC1 / I2C / MIBSPI5                  | MIBADC1 G2 / I2C transmit / MIBSPI5[5]                                           | DMAREQ[11]  |
| RTI / MIBSPI1 / MIBSPI3                  | RTI DMAREQ0 / MIBSPI1[6] / MIBSPI3[6]                                            | DMAREQ[12]  |
| RTI / MIBSPI1 / MIBSPI3                  | RTI DMAREQ1 / MIBSPI1[7] / MIBSPI3[7]                                            | DMAREQ[13]  |
| MIBSPI3 / USB Device / MibADC2 / MIBSPI5 | MIBSPI3[1] <sup>(4)</sup> / USB_FUNC.DMATXREQ_ON[0] / MibADC2 event / MIBSPI5[6] | DMAREQ[14]  |
| MIBSPI3 / USB Device / MIBSPI5           | MIBSPI3[0] <sup>(2)</sup> / USB_FUNC.DMARXREQ_ON[0] / MIBSPI5[7]                 | DMAREQ[15]  |
| MIBSPI1 / MIBSPI3 / DCAN1 / MibADC2      | MIBSPI1[8] / MIBSPI3[8] / DCAN1 IF3 / MibADC2 G1                                 | DMAREQ[16]  |
| MIBSPI1 / MIBSPI3 / DCAN3 / MibADC2      | MIBSPI1[9] / MIBSPI3[9] / DCAN3 IF1 / MibADC2 G2                                 | DMAREQ[17]  |
| RTI / USB Device / MIBSPI5               | RTI DMAREQ2 / USB_FUNC.DMATXREQ_ON[1] / MIBSPI5[8]                               | DMAREQ[18]  |
| RTI / USB Device / MIBSPI5               | RTI DMAREQ3 / USB_FUNC.DMARXREQ_ON[1] / MIBSPI5[9]                               | DMAREQ[19]  |
| N2HET1 / N2HET2 / DCAN3                  | N2HET1 DMAREQ[4] / N2HET2 DMAREQ[4] / DCAN3 IF2                                  | DMAREQ[20]  |
| N2HET1 / N2HET2 / DCAN3                  | N2HET1 DMAREQ[5] / N2HET2 DMAREQ[5] / DCAN3 IF3                                  | DMAREQ[21]  |
| MIBSPI1 / MIBSPI3 / MIBSPI5              | MIBSPI1[10] / MIBSPI3[10] / MIBSPI5[10]                                          | DMAREQ[22]  |
| MIBSPI1 / MIBSPI3 / MIBSPI5              | MIBSPI1[11] / MIBSPI3[11] / MIBSPI5[11]                                          | DMAREQ[23]  |
| N2HET1 / N2HET2 / SPI4 / MIBSPI5         | N2HET1 DMAREQ[6] / N2HET2 DMAREQ[6] / SPI4 receive / MIBSPI5[12]                 | DMAREQ[24]  |
| N2HET1 / N2HET2 / SPI4 / MIBSPI5         | N2HET1 DMAREQ[7] / N2HET2 DMAREQ[7] / SPI4 transmit / MIBSPI5[13]                | DMAREQ[25]  |
| CRC / MIBSPI1 / MIBSPI3                  | CRC DMAREQ[0] / MIBSPI1[12] / MIBSPI3[12]                                        | DMAREQ[26]  |
| CRC / MIBSPI1 / MIBSPI3                  | CRC DMAREQ[1] / MIBSPI1[13] / MIBSPI3[13]                                        | DMAREQ[27]  |
| LIN / USB Device / MIBSPI5               | LIN receive / USB_FUNC.DMATXREQ_ON[2] / MIBSPI5[14]                              | DMAREQ[28]  |
| LIN / USB Device / MIBSPI5               | LIN transmit / USB_FUNC.DMARXREQ_ON[2] / MIBSPI5[15]                             | DMAREQ[29]  |
| MIBSPI1 / MIBSPI3 / SCI / MIBSPI5        | MIBSPI1[14] / MIBSPI3[14] / SCI receive / MIBSPI5[1] <sup>(3)</sup>              | DMAREQ[30]  |
| MIBSPI1 / MIBSPI3 / SCI / MIBSPI5        | MIBSPI1[15] / MIBSPI3[15] / SCI transmit / MIBSPI5[0] <sup>(4)</sup>             | DMAREQ[31]  |

<sup>(3)</sup> SPI1, SPI3, SPI5 receive in standard SPI mode

<sup>(4)</sup> SPI1, SPI3, SPI5 transmit in standard SPI mode

### 16.2.8 Auto-Initiation

When Auto-initiation Mode (AIM) bit of Channel Control Register ([Section 16.3.2.4](#)) is enabled for a channel and the channel is triggered by a software request for a block transfer, the channel will restart again using the same channel information stored at the respective control packet after one block transfer is completed. In the case of Hardware Request, the channel needs to be retriggered each time after a block is complete even if auto-initiation is enabled.

### 16.2.9 Interrupts

Each channel can be configured to generate interrupts on several transfer conditions:

- Frame transfer complete (FTC) interrupt: an interrupt is issued after the last element of a frame has been transferred.
- Last frame transfer started (LFS) interrupt: an interrupt is issued before the first element of the last frame of a block transfer has started.
- First half of block complete (HBC) interrupt: an interrupt is issued if more than half of the block is transferred.
  - If the number of frames  $n$  is odd, then the HBC interrupt is generated at the end of the frame when  $(n+1) / 2$  number of frames are left in the block.
  - If the number of frames  $n$  is even, then the HBC interrupt is generated at the end of the frame after  $n/2$  number of frames are left in the block.
- Block transfer complete (BTC) interrupt: an interrupt is issued after the last element of the last frame has been transferred.
- External imprecise error on read: an interrupt can be issued when a bus error (Illegal transaction with ok response) is detected. The imprecise read error is connected to the ESM module.
- External imprecise error on write: an interrupt can be issued when a bus error (Illegal transaction with ok response) is detected. The imprecise write error is connected to the ESM module.
- Memory Protection Unit error (MPU): an interrupt is issued when the DMA detects that the access falls outside of a memory region programmed in the MPU registers of the DMA. The MPU interrupt is connected to the ESM module.
- Parity error (PAR): an interrupt is issued when the DMA detects a parity error when reading one of the control packets. The PAR interrupt is connected to the ESM module.

The DMA outputs 5 interrupt lines for control packet handling, a parity interrupt and a memory protection interrupt ([Figure 16-14](#)). Each type of transfer interrupt condition is grouped together. For example, all block-transfer complete interrupts that are routed to a port are combined (ORed). The channel that caused the interrupt is given in the corresponding interrupt channel offset register. Priority between interrupts among the same interrupt type is resolved by a fixed priority scheme. Priority between different interrupt types is resolved in the Vector Interrupt Manager. [Figure 16-15](#) explains the Frame Transfer Complete Interrupt structure in detail.

---

**NOTE:** Each Channel Specific interrupts in DMA module are routed towards Group A or B to support two different CPUs individually. For devices with Single CPU or Dual CPU where both CPUs are running same code in delayed lock-step as safety feature:

Group A - Interrupts (FTC, LFS, HBC, and BTC) are routed to the ARM CPU.

Group B - Interrupts (FTC, LFS, HBC, and BTC) are not routed out.

User software should configure only Group A interrupts.

---

Figure 16-14. DMA Interrupts

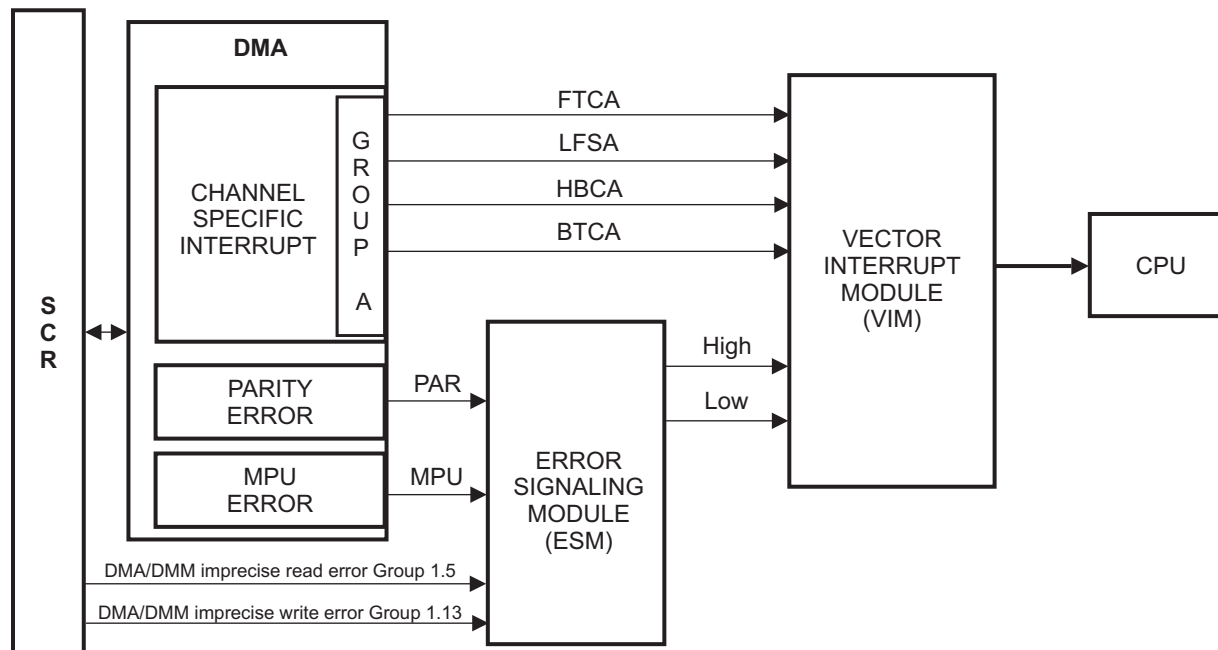
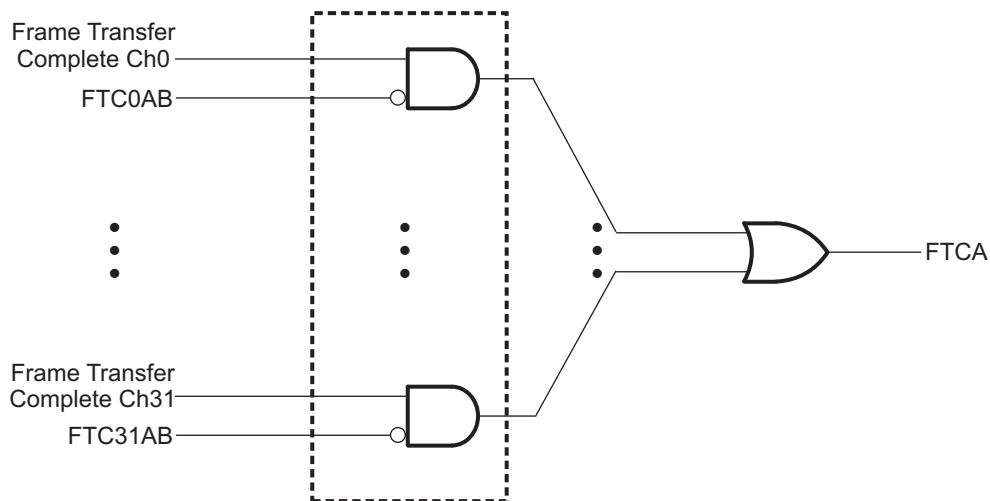


Figure 16-15. Detailed Interrupt Structure (Frame Transfer Complete Path)



This figure is applicable for the HBC, LFS, and BTC interrupt.

### 16.2.10 Debugging

The DMA supports four different behaviors in suspend mode. These behaviors can be configured by the user as per the application requirement.

- Immediate stop at a DMA channel arbitration boundary. Please refer to [Table 16-3](#) and [Table 16-4](#) for arbitration boundary definition.
- Finish current frame transfer and continue after suspend ends.
- Finish current block transfer and continue after suspend ends.
- Ignore the suspend. The DMA continues to be operational as in functional mode when debug mode is active.

When the DMA controller enters suspend mode, it continues to sample incoming hardware DMA requests, but the Channel Pending Register ([Section 16.3.1.2](#)) is frozen from being updated. After the suspend ends, all new requests that were received during suspend mode are reflected in the Channel Pending Register ([Section 16.3.1.2](#)).

Except when the DMA controller is configured to ignore suspend mode, no channel arbitration is performed during suspend mode. The current channel under which suspend mode was entered will finish its entire frame or block-transfer after suspend mode ends, depending how the debug option was chosen.

To facilitate debugging, a Watch Point Register ([Section 16.3.1.47](#)) and a Watch Mask Register ([Section 16.3.1.48](#)) are used. The watch point register together with the watch mask register can be configured to watch for a unique address or a range of addresses. When the condition to watch is true, the DMA freezes its state and generates a debug request signal to the host CPU so the state of the DMA can be examined.

### 16.2.11 Power Management

The DMA offers two power-management modes: run and sleep. In run mode, the DMA is fully operational.

The sleep mode shuts down the DMA if no pending channels are waiting to be serviced. If a DMA request is received or a software request is generated by the user software, then the DMA wakes up immediately.

The sleep mode may be used to optimize the DMA module power consumption.

When the system module issues a global low power mode request, the DMA will respond to the system module with an acknowledge if no DMA requests are pending.

---

**NOTE:** When the DMA is in global low power mode, the clock is stopped and therefore it cannot detect any DMA request. The device must be woken up before a peripheral can generate a DMA request.

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### 16.2.12 FIFO Buffer

DMA FIFO is 4 levels deep and 64-bit wide (can hold up to 4 x 64-bits of data). They are used for Data packing and unpacking.

The DMA FIFO has two states:

- **EMPTY** : The FIFO contains no data.
- **FULL** : The FIFO is filled or the element count has reached zero; the read operation has to be stopped.

DMA channels can only be switched when the FIFO is empty. This also implies that arbitration between channels is done when the FIFO is empty.

The FIFO buffer may be bypassed through the use of the bypass feature in the port control register; see Port Control Register ([Section 16.3.1.44](#)) for register details. Writing 1 to this bit limits the FIFO depth to the size of one element. That means if the read element size is equal to or larger than the write element size, after one element is read the write out to the destination starts. Otherwise, the write out to the destination starts after enough reads have completed to do one write of the write element size. This feature is particularly useful to minimize switching latency in-between channels. When bypass mode is enabled, the DMA performs minimal bus cycles on AHB bus. In addition, the bypass feature allows arbitration between channels that can be carried out at a source element granularity.

However, it has to be considered that while in bypass mode, the DMA controller does not make optimal use of the bus bandwidth. Since the read and write element sizes can be different, then the number of read and write transactions will be different. [Table 16-3](#) and [Table 16-4](#) show a comparison between the number of read and write transactions performed by the DMA controller from one channel to another before arbitration in non-bypass and bypass mode.

**Table 16-3. Maximum Number of DMA Transactions per Channel in Non-Bypass Mode**

|                   | Write Element Size | 8 bit  |         | 16 bit |         | 32 bit |         | 64 bit |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
| Read Element Size | 8 bit              | 4 read | 4 write | 4 read | 2 write | 4 read | 1 write | 8 read | 1 write |
|                   | 16 bit             | 2 read | 4 write | 4 read | 4 write | 4 read | 2 write | 4 read | 1 write |
|                   | 32 bit             | 1 read | 4 write | 2 read | 4 write | 4 read | 4 write | 4 read | 2 write |
|                   | 64 bit             | 1 read | 8 write | 1 read | 4 write | 2 read | 4 write | 4 read | 4 write |

**Table 16-4. Maximum Number of DMA Transactions per Channel in Bypass Mode**

|                   | Write Element Size | 8 bit  |         | 16 bit |         | 32 bit |         | 64 bit |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
| Read Element Size | 8 bit              | 1 read | 1 write | 2 read | 1 write | 4 read | 1 write | 8 read | 1 write |
|                   | 16 bit             | 1 read | 2 write | 1 read | 1 write | 2 read | 1 write | 4 read | 1 write |
|                   | 32 bit             | 1 read | 4 write | 1 read | 2 write | 1 read | 1 write | 2 read | 1 write |
|                   | 64 bit             | 1 read | 8 write | 1 read | 4 write | 1 read | 2 write | 1 read | 1 write |

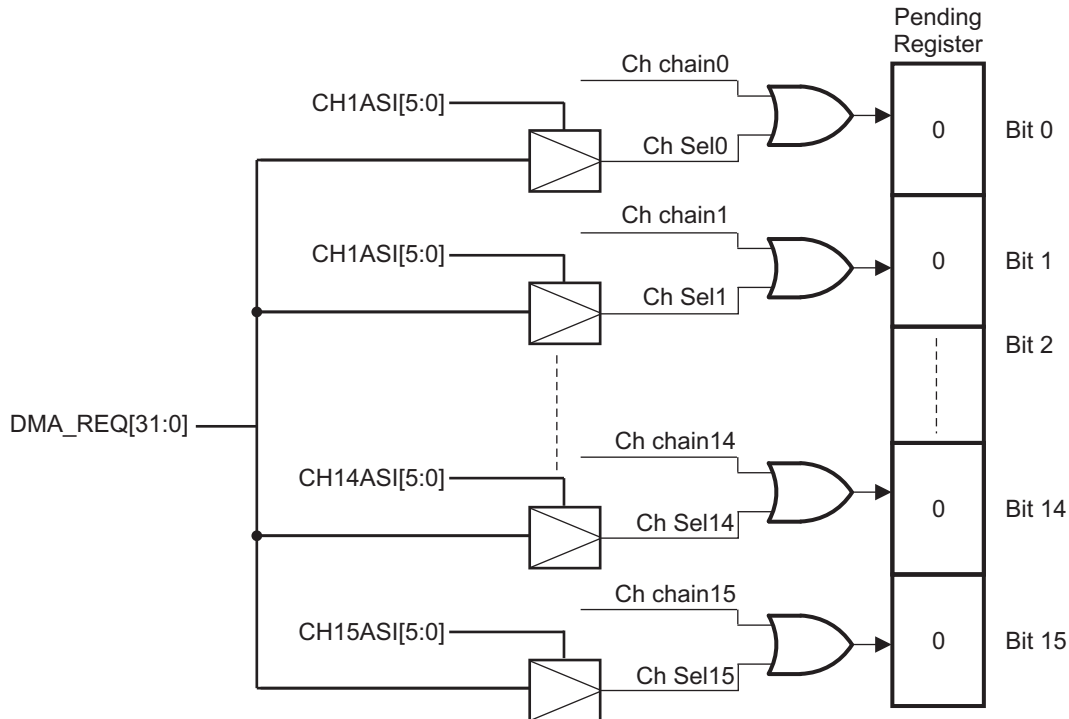
### 16.2.13 Channel Chaining

Channel chaining is used to trigger a single or multiple channels with out an external DMA request. This is possible by chaining one control packet to other. Chain[5:0] field of the Channel Control Register ([Section 16.3.2.4](#)) is used to program the chaining control packet. Chained control packets follow arbitration rules within the pending register. For example if CH1, CH2, CH4, CH5 are triggered together and CH3 is chained with CH1. The order of channels serviced in spite of chaining will be CH1 -> CH2 -> CH3 -> CH4 -> CH5.

In order to setup up channel chain feature, the Channel Control Register ([Section 16.3.2.4](#)) needs to be enabled for all chained channels before triggering first DMA request.

[Figure 16-16](#) illustrates how internally chained request is generated after completing the required transfers and stored in pending register. In this example CH1 is Chained to CH0. When CH0 is triggered CH1 is captured as pending in the Channel Pending Register ([Section 16.3.1.2](#)) even when it is not triggered.

**Figure 16-16. Example of Channel Chaining**



### 16.2.14 Memory Protection

The DMA controller is capable of access to the full address range of the device. The protection mechanism allows the protection of up to four memory regions to restrict accesses to those address ranges. This will allow the application to protect critical application data from unintentionally being accessed by the DMA controller.

#### 16.2.14.1 Protection Mechanism

The memory protection mechanism consists of the access privilege for a given memory region, the start and end address for the region, and notification of an access violation for the protected region.

Each region to be protected is configured by software by writing the start address and end address for each region into the DMA Memory Protection Registers, DMAMPRxS and DMAMPRxE. The definition of these registers can be found starting at [Section 16.3.1.54](#). Any region in the valid address space can be protected from inappropriate accesses.

The access privileges can be set to one of four permission settings:

- Full access
- Read only access
- Write only access
- No access

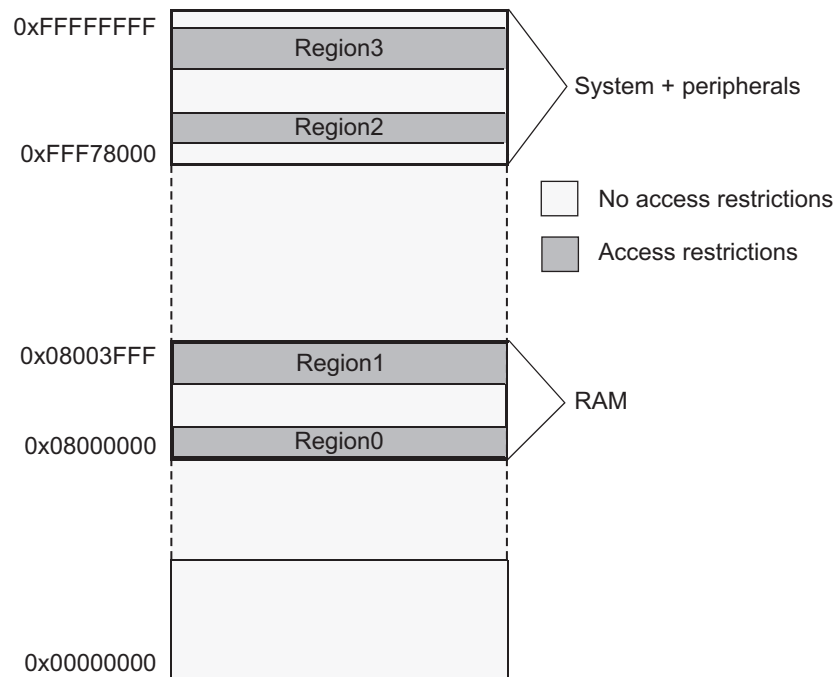
The permissions for a given region are selected by writing the appropriate values in the DMA Memory Protection Control Register ([Section 16.3.1.54](#)).

**NOTE:** If the regions defined by the start and end addresses overlap, the region defined first in the register space determines the access privilege. For example, if region 0 and region 1 overlap, the access permissions defined for region 0 will take precedence since region 0 registers are before region 1.

In a case where a memory protection violation occurs, a flag will be set and an interrupt will be generated, if interrupts are enabled. The DMA Memory Protection Status Register ([Section 16.3.1.55](#)) contains the status flags for the memory protection mechanism, and the DMA Memory Protection Control Register ([Section 16.3.1.54](#)) contains the interrupt enable bits. Upon detection of the memory protection violation, the DMA Channel that caused the violation will be stopped and the next available DMA channel will be serviced.

Figure 16-17 illustrates a protection mechanism.

**Figure 16-17. Example of Protection Mechanism**



### 16.2.15 Parity Checking

Parity checking is implemented using parity on a per-byte basis for DMA Control Packets in the RAM. Checking for even or odd parity can be programmed by a 4-bit key located in the system module that controls the parity configuration on a global basis. This ensures that all modules using parity are acting in the same manner. The default setup after reset is odd parity.

In addition, parity checking can be enabled and disabled within the module by a 4-bit key. The key is located in the Parity Control Register ([Section 16.3.1.52](#)).

During a read access, regardless if it was read by the DMA state machine or another master (CPU), the parity is calculated based on the data read from the RAM and compared with the good parity value stored in the parity bits. If any word fails the parity check, then a parity error interrupt is generated. The address that generated the error is detected and is captured for host system debugging in the DMA Parity Error Address Register ([Section 16.3.1.53](#)). The address is frozen from being updated until it is read by the bus master.

Additional error handling is dependent on the requestor.

- DMA reading from a control packet RAM: The transmission requested by DMA request will not take place.
- CPU reading from the control packet RAM: The data will be retrieved by the CPU and a parity error interrupt will be generated.

In both cases, the control packet will be left active or the DMA will be switched off dependent on the ERRR bit in the Parity Control Register ([Section 16.3.1.52](#)).

### 16.2.16 Parity Testing

The parity RAM is accessible to allow manually inserting faults so that the parity checking feature can be tested. Test mode is entered by asserting the TEST bit in the Parity Control Register ([Section 16.3.1.52](#)). Once the bit is set, the parity bits are mapped to the control packet RAM starting address A00h.

**NOTE:** When in test mode, no parity checking will be done when reading from parity memory, but parity checking will be performed on the normal memory.

Each byte in Control Packet RAM has its own parity bit in the Control Packet Parity RAM as shown in [Table 16-5](#), [Table 16-6](#), and [Table 16-7](#). P0 is the parity bit for byte 0, P1 is the parity bit for byte 1 and so on.

Each byte in the control packet RAM has its own parity bit in the control packet parity RAM as shown in [Table 16-5](#) and [Table 16-6](#).

**Table 16-5. Control Packet RAM**

| Bit   | 31      | 24 | 23      | 16 | 15      | 8 | 7       | 0 |
|-------|---------|----|---------|----|---------|---|---------|---|
| Word0 | Byte 0  |    | Byte 1  |    | Byte 2  |   | Byte 3  |   |
| Word1 | Byte 4  |    | Byte 5  |    | Byte 6  |   | Byte 7  |   |
| Word2 | Byte 8  |    | Byte 9  |    | Byte 10 |   | Byte 11 |   |
| Word3 | Byte 12 |    | Byte 13 |    | Byte 14 |   | Byte 15 |   |

**Table 16-6. Control Packet RAM**

| Bit | 127    | 96 | 95     | 64 | 63     | 32 | 31     | 0 |
|-----|--------|----|--------|----|--------|----|--------|---|
|     | Word 3 |    | Word 2 |    | Word 1 |    | Word 0 |   |

**Table 16-7. Parity RAM**

| Bit | 15  | 14  | 13  | 12  | 11  | 10  | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
|-----|-----|-----|-----|-----|-----|-----|----|----|----|----|----|----|----|----|----|----|
|     | P15 | P14 | P13 | P12 | P11 | P10 | P9 | P8 | P7 | P6 | P5 | P4 | P3 | P2 | P1 | P0 |

### 16.2.17 Initializing RAM with Parity

After power up, the RAM content including the parity bit cannot be guaranteed. To avoid parity failures when reading RAM, the RAM has to be initialized. The RAM can be initialized by writing known values into it. When the known value is written, the corresponding parity bit will be automatically calculated and updated.

Another possibility to initialize the memory is to follow the Auto-Initialization of On-Chip SRAM Modules subsection in the *Architecture* chapter. The RAM will be initialized to 0. Depending on the even/odd parity selection, the parity bit will be calculated accordingly.

To allow for parity calculation during initialization, the parity functionality has to be enabled as discussed in [Section 16.2.15](#).

## 16.3 Control Registers and Control Packets

The DMA control registers are summarized in [Table 16-8](#). The base address for the control registers is FFFF F000h. The control packets are summarized in [Table 16-9](#). The base address for the control packets is FFF8 0000h. Each register begins on a word boundary. All registers and control packets are accessible in 8, 16, and 32 bit.

**NOTE:** The register definitions are given for a full DMA module configuration (32 channels, 64 requests, 2 Ports, Dual CPU support). Writes and Reads of bits pertaining to features not included in the DMA implementation as defined in the device-specific data manual are possible without error; however, they will have no affect on device operation.

**Table 16-8. DMA Control Registers**

| Offset | Acronym    | Register Description                           | Section                           |
|--------|------------|------------------------------------------------|-----------------------------------|
| 00h    | GCTRL      | Global Control Register                        | <a href="#">Section 16.3.1.1</a>  |
| 04h    | PEND       | Channel Pending Register                       | <a href="#">Section 16.3.1.2</a>  |
| 0Ch    | DMASTAT    | DMA Status Register                            | <a href="#">Section 16.3.1.3</a>  |
| 14h    | HWCHENAS   | HW Channel Enable Set and Status Register      | <a href="#">Section 16.3.1.4</a>  |
| 1Ch    | HWCHENAR   | HW Channel Enable Reset and Status Register    | <a href="#">Section 16.3.1.5</a>  |
| 24h    | SWCHENAS   | SW Channel Enable Set and Status Register      | <a href="#">Section 16.3.1.6</a>  |
| 2Ch    | SWCHENAR   | SW Channel Enable Reset and Status Register    | <a href="#">Section 16.3.1.7</a>  |
| 34h    | CHPRIOS    | Channel Priority Set Register                  | <a href="#">Section 16.3.1.8</a>  |
| 3Ch    | CHPRIOR    | Channel Priority Reset Register                | <a href="#">Section 16.3.1.9</a>  |
| 44h    | GCHIENAS   | Global Channel Interrupt Enable Set Register   | <a href="#">Section 16.3.1.10</a> |
| 4Ch    | GCHIENAR   | Global Channel Interrupt Enable Reset Register | <a href="#">Section 16.3.1.11</a> |
| 54h    | DREQASI0   | DMA Request Assignment Register 0              | <a href="#">Section 16.3.1.12</a> |
| 58h    | DREQASI1   | DMA Request Assignment Register 1              | <a href="#">Section 16.3.1.13</a> |
| 5Ch    | DREQASI2   | DMA Request Assignment Register 2              | <a href="#">Section 16.3.1.14</a> |
| 60h    | DREQASI3   | DMA Request Assignment Register 3              | <a href="#">Section 16.3.1.15</a> |
| 94h    | PAR0       | Port Assignment Register 0                     | <a href="#">Section 16.3.1.16</a> |
| 98h    | PAR1       | Port Assignment Register 1                     | <a href="#">Section 16.3.1.17</a> |
| B4h    | FTCMAP     | FTC Interrupt Mapping Register                 | <a href="#">Section 16.3.1.18</a> |
| BCh    | LFSMAP     | LFS Interrupt Mapping Register                 | <a href="#">Section 16.3.1.19</a> |
| C4h    | HBCMAP     | HBC Interrupt Mapping Register                 | <a href="#">Section 16.3.1.20</a> |
| CCh    | BTCMAP     | BTC Interrupt Mapping Register                 | <a href="#">Section 16.3.1.21</a> |
| DCh    | FTCINTENAS | FTC Interrupt Enable Set Register              | <a href="#">Section 16.3.1.22</a> |
| E4h    | FTCINTENAR | FTC Interrupt Enable Reset Register            | <a href="#">Section 16.3.1.23</a> |
| ECh    | LFSINTENAS | LFS Interrupt Enable Set Register              | <a href="#">Section 16.3.1.24</a> |
| F4h    | LFSINTENAR | LFS Interrupt Enable Reset Register            | <a href="#">Section 16.3.1.25</a> |
| FCh    | HBCINTENAS | HBC Interrupt Enable Set Register              | <a href="#">Section 16.3.1.26</a> |
| 104h   | HBCINTENAR | HBC Interrupt Enable Reset Register            | <a href="#">Section 16.3.1.27</a> |
| 10Ch   | BTCINTENAS | BTC Interrupt Enable Set Register              | <a href="#">Section 16.3.1.28</a> |
| 114h   | BTCINTENAR | BTC Interrupt Enable Reset Register            | <a href="#">Section 16.3.1.29</a> |
| 11Ch   | GINTFLAG   | Global Interrupt Flag Register                 | <a href="#">Section 16.3.1.30</a> |
| 124h   | FTCFLAG    | FTC Interrupt Flag Register                    | <a href="#">Section 16.3.1.31</a> |
| 12Ch   | LFSFLAG    | LFS Interrupt Flag Register                    | <a href="#">Section 16.3.1.32</a> |
| 134h   | HBCFLAG    | HBC Interrupt Flag Register                    | <a href="#">Section 16.3.1.33</a> |
| 13Ch   | BTCFLAG    | BTC Interrupt Flag Register                    | <a href="#">Section 16.3.1.34</a> |
| 144h   | BERFLAG    | BER Interrupt Flag Register                    | <a href="#">Section 16.3.1.35</a> |
| 14Ch   | FTCAOFFSET | FTCA Interrupt Channel Offset Register         | <a href="#">Section 16.3.1.36</a> |
| 150h   | LFSAOFFSET | LFSA Interrupt Channel Offset Register         | <a href="#">Section 16.3.1.37</a> |

**Table 16-8. DMA Control Registers (continued)**

| Offset | Acronym    | Register Description                                  | Section                           |
|--------|------------|-------------------------------------------------------|-----------------------------------|
| 154h   | HBCAOFFSET | HBCA Interrupt Channel Offset Register                | <a href="#">Section 16.3.1.38</a> |
| 158h   | BTCAOFFSET | BTCA Interrupt Channel Offset Register                | <a href="#">Section 16.3.1.39</a> |
| 160h   | FTCBOFFSET | FTCB Interrupt Channel Offset Register                | <a href="#">Section 16.3.1.40</a> |
| 164h   | LFSBOFFSET | LFSB Interrupt Channel Offset Register                | <a href="#">Section 16.3.1.41</a> |
| 168h   | HBCBOFFSET | HBCB Interrupt Channel Offset Register                | <a href="#">Section 16.3.1.42</a> |
| 16Ch   | BTCBOFFSET | BTCB Interrupt Channel Offset Register                | <a href="#">Section 16.3.1.43</a> |
| 178h   | PTCRL      | Port Control Register                                 | <a href="#">Section 16.3.1.44</a> |
| 17Ch   | RTCTRL     | RAM Test Control Register                             | <a href="#">Section 16.3.1.45</a> |
| 180h   | DCTRL      | Debug Control Register                                | <a href="#">Section 16.3.1.46</a> |
| 184h   | WPR        | Watch Point Register                                  | <a href="#">Section 16.3.1.47</a> |
| 188h   | WMR        | Watch Mask Register                                   | <a href="#">Section 16.3.1.48</a> |
| 198h   | PBACSADDR  | Port B Active Channel Source Address Register         | <a href="#">Section 16.3.1.49</a> |
| 19Ch   | PBACDADDR  | Port B Active Channel Destination Address Register    | <a href="#">Section 16.3.1.50</a> |
| 1A0h   | PBACTC     | Port B Active Channel Transfer Count Register         | <a href="#">Section 16.3.1.51</a> |
| 1A8h   | DMAPCR     | Parity Control Register                               | <a href="#">Section 16.3.1.52</a> |
| 1ACh   | DMAPAR     | DMA Parity Error Address Register                     | <a href="#">Section 16.3.1.53</a> |
| 1B0h   | DMAMPCTRL  | DMA Memory Protection Control Register                | <a href="#">Section 16.3.1.54</a> |
| 1B4h   | DMAMPST    | DMA Memory Protection Status Register                 | <a href="#">Section 16.3.1.55</a> |
| 1B8h   | DMAMPR0S   | DMA Memory Protection Region 0 Start Address Register | <a href="#">Section 16.3.1.56</a> |
| 1BCh   | DMAMPR0E   | DMA Memory Protection Region 0 End Address Register   | <a href="#">Section 16.3.1.57</a> |
| 1C0h   | DMAMPR1S   | DMA Memory Protection Region 1 Start Address Register | <a href="#">Section 16.3.1.58</a> |
| 1C4h   | DMAMPR1E   | DMA Memory Protection Region 1 End Address Register   | <a href="#">Section 16.3.1.59</a> |
| 1C8h   | DMAMPR2S   | DMA Memory Protection Region 2 Start Address Register | <a href="#">Section 16.3.1.60</a> |
| 1CCh   | DMAMPR2E   | DMA Memory Protection Region 2 End Address Register   | <a href="#">Section 16.3.1.61</a> |
| 1D0h   | DMAMPR3S   | DMA Memory Protection Region 3 Start Address Register | <a href="#">Section 16.3.1.62</a> |
| 1D4h   | DMAMPR3E   | DMA Memory Protection Region 3 End Address Register   | <a href="#">Section 16.3.1.63</a> |

**Table 16-9. Control Packet Memory Map**

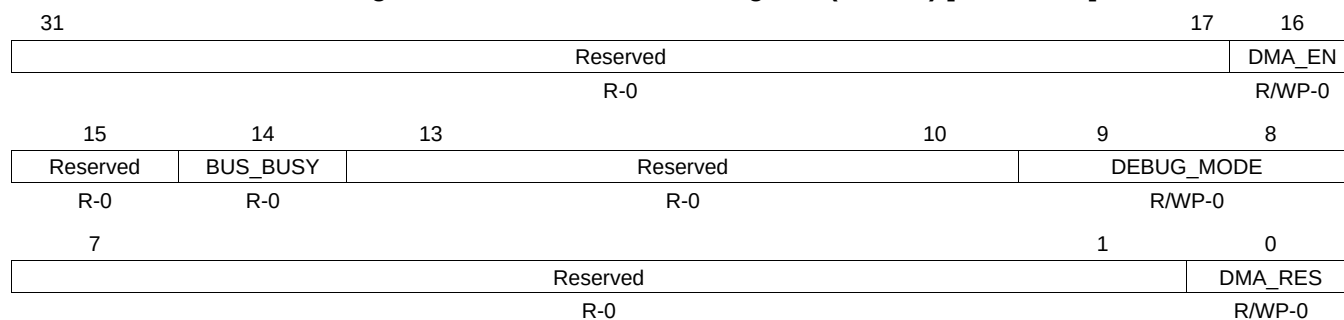
| Offset                          | Acronym | Register Description                 | Section                          |
|---------------------------------|---------|--------------------------------------|----------------------------------|
| <b>Primary Control Packet 0</b> |         |                                      |                                  |
| 00h                             | ISADDR  | Initial Source Address Register      | <a href="#">Section 16.3.2.1</a> |
| 04h                             | IDADDR  | Initial Destination Address Register | <a href="#">Section 16.3.2.2</a> |
| 08h                             | ITCOUNT | Initial Transfer Count Register      | <a href="#">Section 16.3.2.3</a> |
| 10h                             | CHCTRL  | Channel Control Register             | <a href="#">Section 16.3.2.4</a> |
| 14h                             | EIOFF   | Element Index Offset Register        | <a href="#">Section 16.3.2.5</a> |
| 18h                             | FIOFF   | Frame Index Offset Register          | <a href="#">Section 16.3.2.6</a> |
| <b>Working Control Packet 0</b> |         |                                      |                                  |
| 800h                            | CSADDR  | Current Source Address Register      | <a href="#">Section 16.3.2.7</a> |
| 804h                            | CDADDR  | Current Destination Address Register | <a href="#">Section 16.3.2.8</a> |
| 808h                            | CTCOUNT | Current Transfer Count Register      | <a href="#">Section 16.3.2.9</a> |

### 16.3.1 Global Configuration Registers

These registers control the overall behavior of the DMA controller.

#### 16.3.1.1 Global Control Register (GCTRL)

**Figure 16-18. Global Control Register (GCTRL) [offset = 00]**



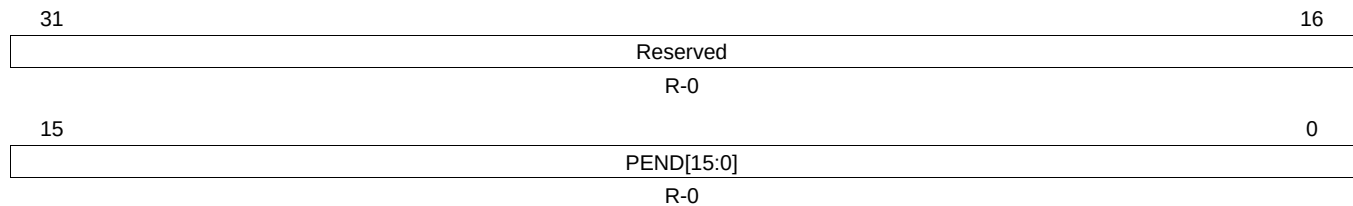
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

### Table 16-10. Global Control Register (GCTRL) Field Descriptions

| Bit   | Field      | Value | Description                                                                                                                                                                                                                                                                                                     |
|-------|------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-17 | Reserved   | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                          |
| 16    | DMA_EN     |       | DMA enable bit. The configuration registers and channel control packets should be setup first before DMA_EN bit is set to 1 to prevent state machines from carrying out bus transactions. If DMA_EN bit is cleared in the middle of an bus transaction, the state machine will stop at an arbitration boundary. |
|       |            | 0     | The DMA is disabled.                                                                                                                                                                                                                                                                                            |
|       |            | 1     | The DMA is enabled.                                                                                                                                                                                                                                                                                             |
| 15    | Reserved   | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                          |
| 14    | BUS_BUSY   |       | This bit indicates status of DMA external AHB bus status.                                                                                                                                                                                                                                                       |
|       |            | 0     | DMAs external bus is not busy in data transfers.                                                                                                                                                                                                                                                                |
|       |            | 1     | DMAs external bus is busy in data transfers.                                                                                                                                                                                                                                                                    |
| 13-10 | Reserved   | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                          |
| 9-8   | DEBUG_MODE |       | Debug Mode.                                                                                                                                                                                                                                                                                                     |
|       |            | 0     | Ignore suspend.                                                                                                                                                                                                                                                                                                 |
|       |            | 1h    | Finish current block transfer.                                                                                                                                                                                                                                                                                  |
|       |            | 2h    | Finish current frame transfer.                                                                                                                                                                                                                                                                                  |
|       |            | 3h    | Immediately stop at an DMA arbitration boundary and continue after suspend.                                                                                                                                                                                                                                     |
| 7-1   | Reserved   | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                          |
| 0     | DMA_RES    |       | DMA software reset                                                                                                                                                                                                                                                                                              |
|       |            |       | <b>Note: In the event a DMA slave does not respond, the DMA module will respond to the software reset upon reaching an arbitration boundary.</b>                                                                                                                                                                |
|       |            | 0     | Read: Software reset is disabled.<br>Write: No effect.                                                                                                                                                                                                                                                          |
|       |            | 1     | Read and write: The DMA state machine and all control registers are in software reset. Control packets are not reset when DMA software reset is active.                                                                                                                                                         |

### 16.3.1.2 Channel Pending Register (PEND)

**Figure 16-19. Channel Pending Register (PEND) [offset = 04h]**



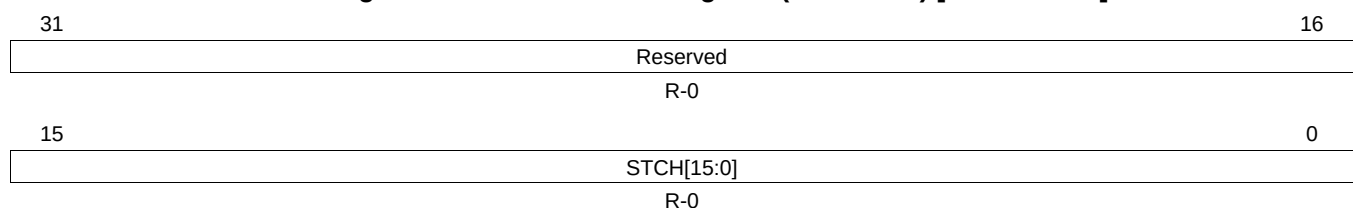
LEGEND: R = Read only; -n = value after reset

**Table 16-11. Channel Pending Register (PEND) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|----------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 15-0  | PEND[n]  |       | Channel pending register. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. Reading from PEND gives the channel pending information if the channel was initiated by SW or HW. Once set, it remains set even if the corresponding channel is disabled via HWCHENA or SWCHENA. The pending bit is automatically cleared for the following conditions: <ul style="list-style-type: none"> <li>At the end of a frame or a block transfer depending on how the channel is triggered as programmed in the TTYPE bit field of CHCTRL.</li> <li>The control packet is modified after the pending bit is set.</li> <li>An AHB bus error occurs.</li> </ul> |
|       |          | 0     | The corresponding channel is inactive.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|       |          | 1     | The corresponding channel is pending and is waiting for service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

### 16.3.1.3 DMA Status Register (DMASTAT)

**Figure 16-20. DMA Status Register (DMASTAT) [offset = 0Ch]**



LEGEND: R = Read only; -n = value after reset

**Table 16-12. DMA Status Register (DMASTAT) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                                                                                                           |
|-------|----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                |
| 15-0  | STCH[n]  |       | Status of DMA channels. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.                                                                                                                                                    |
|       |          | 0     | The channel is inactive.                                                                                                                                                                                                                              |
|       |          | 1     | The channel is active; that is, the channel is currently in the DMA's execution queue.<br><b>Note: The status of a channel currently in DMA's execution queue remains active even if emulation mode is entered or DMA is disabled via DMA_EN bit.</b> |

#### 16.3.1.4 HW Channel Enable Set and Status Register (HWCHENAS)

**Figure 16-21. HW Channel Enable Set and Status Register (HWCHENAS) [offset = 14h]**

|               |    |
|---------------|----|
| 31            | 16 |
| Reserved      |    |
| R-0           |    |
| 15            | 0  |
| HWCHENA[15:0] |    |
| R/WP-0        |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-13. HW Channel Enable Set and Status Register (HWCHENAS) Field Descriptions**

| Bit   | Field      | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved   | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 15-0  | HWCHENA[n] |       | Hardware channel enable bit. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. An active hardware DMA request cannot initiate a DMA transfer unless the corresponding hardware enable bit is set.<br><br>The corresponding hardware enable bit is cleared automatically for the following conditions: <ul style="list-style-type: none"> <li>At the end of a block transfer if the auto-initiation bit AIM (see CHCTRL) is not active.</li> <li>If an AHB bus error is detected for an active channel.</li> </ul> Reading from HWCHENAS gives the status (enabled/disabled) of channels 0 through 15. |
|       |            | 0     | The corresponding channel is disabled for hardware triggering.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|       |            | 1     | The corresponding channel is enabled for hardware triggering.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

#### 16.3.1.5 HW Channel Enable Reset and Status Register (HWCHENAR)

**Figure 16-22. HW Channel Enable Reset and Status Register (HWCHENAR) [offset = 1Ch]**

|               |    |
|---------------|----|
| 31            | 16 |
| Reserved      |    |
| R-0           |    |
| 15            | 0  |
| HWCHDIS[15:0] |    |
| R/WP-0        |    |

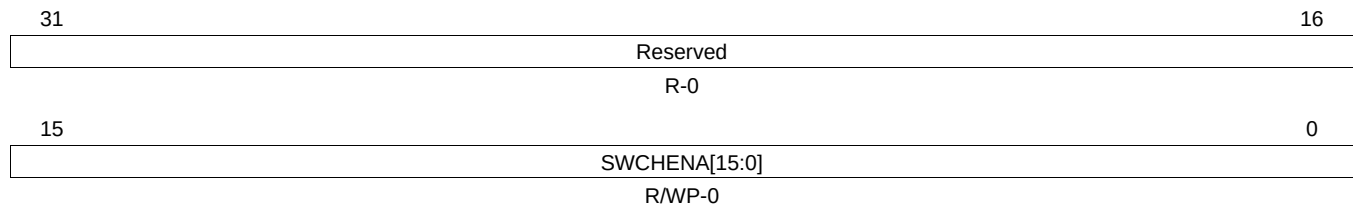
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-14. HW Channel Enable Reset and Status Register (HWCHENAR) Field Descriptions**

| Bit   | Field      | Value | Description                                                                                                    |
|-------|------------|-------|----------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved   | 0     | Reads return 0. Writes have no effect.                                                                         |
| 15-0  | HWCHDIS[n] |       | HW channel disable bit. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.             |
|       |            | 0     | Read: The corresponding channel is disabled for HW triggering.<br>Write: No effect.                            |
|       |            | 1     | Read: The corresponding channel is enabled for HW triggering.<br>Write: The corresponding channel is disabled. |

### 16.3.1.6 SW Channel Enable Set and Status Register (SWCHENAS)

**Figure 16-23. SW Channel Enable Set and Status Register (SWCHENAS) [offset = 24h]**



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-15. SW Channel Enable Set and Status Register (SWCHENAS) Field Descriptions**

| Bit   | Field      | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved   | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 15-0  | SWCHENA[n] |       | <p>SW channel enable bit. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. Writing a 1 to a bit triggers a SW request on the corresponding channel to start a DMA transaction. The corresponding bit is automatically cleared by the following conditions.</p> <ul style="list-style-type: none"> <li>The corresponding bit is cleared after one frame transfer if the TTYPE bit in Channel Control Register (CHCTRL) is programmed for frame transfer.</li> <li>The corresponding bit is cleared after one block transfer if the corresponding TTYPE bit is programmed for block transfer and the auto-initiation bit is not enabled.</li> <li>The control packet is modified after the pending bit is set.</li> <li>The corresponding bit is cleared after one block transfer when TTYPE bit is programmed for blocks transfer and if the corresponding bit in HW channel enable register (HWCHENAS) is enabled. When a channel is enabled for both HW and SW, the state machine will initiate transfers based on the SW first. After one block transfer is complete, the corresponding bit in the SWCHENA register is then cleared. The same channel is serviced again by a HW DMA request.</li> <li>The corresponding bit is cleared if an AHB bus error is detected.</li> </ul> <p>Reading from SWCHENAS gives the status (enabled/disabled) of channels 0 through 15.</p> |
|       |            | 0     | The corresponding channel is not triggered by SW request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|       |            | 1     | The corresponding channel is triggered by SW request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

### 16.3.1.7 SW Channel Enable Reset and Status Register (SWCHENAR)

**Figure 16-24. SW Channel Enable Reset and Status Register (SWCHENAR) [offset = 2Ch]**

|               |    |
|---------------|----|
| 31            | 16 |
| Reserved      |    |
| R-0           |    |
| 15            | 0  |
| SWCHDIS[15:0] |    |
| R/WP-0        |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-16. SW Channel Enable Reset and Status Register (SWCHENAR) Field Descriptions**

| Bit   | Field      | Value | Description                                                                                                                                                                         |
|-------|------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved   | 0     | Reads return 0. Writes have no effect.                                                                                                                                              |
| 15-0  | SWCHDIS[n] | 0     | SW channel disable bit. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.<br>Read: The corresponding channel was not triggered by SW.<br>Write: No effect. |
|       |            | 1     | Read: The corresponding channel was triggered by SW.<br>Write: The corresponding channel is disabled.                                                                               |

### 16.3.1.8 Channel Priority Set Register (CHPRIOS)

**Figure 16-25. Channel Priority Set Register (CHPRIOS) [offset = 34h]**

|           |    |
|-----------|----|
| 31        | 16 |
| Reserved  |    |
| R-0       |    |
| 15        | 0  |
| CPS[15:0] |    |
| R/WP-0    |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-17. Channel Priority Set Register (CHPRIOS) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                                                                                                                                               |
|-------|----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                    |
| 15-0  | CPS[n]   | 0     | Channel priority set bit. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.<br>Writing a 1 to a bit assigns the corresponding channel to the high-priority queue.<br>Read: The corresponding channel is assigned to the low-priority queue.<br>Write: No effect. |
|       |          | 1     | Read and write: The corresponding channel is assigned to high-priority queue.                                                                                                                                                                                                             |

### 16.3.1.9 Channel Priority Reset Register (CHPRIOR)

**Figure 16-26. Channel Priority Reset Register (CHPRIOR) [offset = 3Ch]**

|        |           |    |
|--------|-----------|----|
| 31     | Reserved  | 16 |
| R-0    |           |    |
| 15     | CPR[15:0] | 0  |
| R/WP-0 |           |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-18. Channel Priority Reset Register (CHPRIOR) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                                                                                                                                         |
|-------|----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                              |
| 15-0  | CPR[n]   | 0     | Channel priority reset bit. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. Writing a 1 to a bit assigns the according channel to the low-priority queue.<br>Read: The corresponding channel is assigned to the low-priority queue.<br>Write: No effect. |
|       |          | 1     | Read: The corresponding channel is assigned to the high-priority queue.<br>Write: The corresponding channel is assigned to the low-priority queue.                                                                                                                                  |

### 16.3.1.10 Global Channel Interrupt Enable Set Register (GCHIENAS)

**Figure 16-27. Global Channel Interrupt Enable Set Register (GCHIENAS) [offset = 44h]**

|        |             |    |
|--------|-------------|----|
| 31     | Reserved    | 16 |
| R-0    |             |    |
| 15     | GCHIE[15:0] | 0  |
| R/WP-0 |             |    |

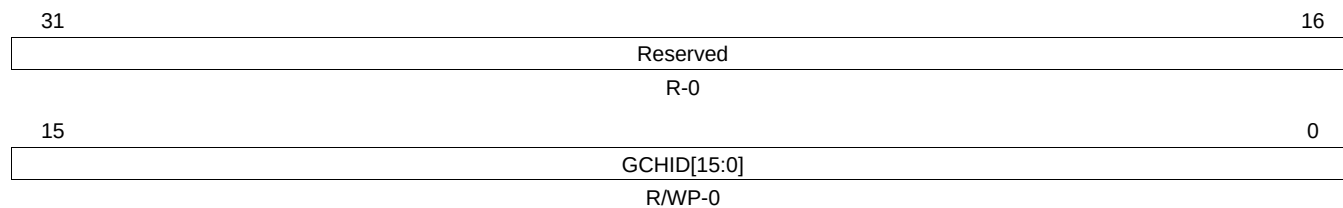
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-19. Global Channel Interrupt Enable Set Register (GCHIENAS) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                                                        |
|-------|----------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                             |
| 15-0  | GCHIE[n] | 0     | Global channel interrupt enable bit. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.<br>Read: The corresponding channel is disabled for interrupt.<br>Write: No effect. |
|       |          | 1     | Read and write: The corresponding channel is enabled for interrupt.                                                                                                                                |

### 16.3.1.11 Global Channel Interrupt Enable Reset Register (GCHIENAR)

**Figure 16-28. Global Channel Interrupt Enable Reset Register (GCHIENAR) [offset = 4Ch]**



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-20. Global Channel Interrupt Enable Reset Register (GCHIENAR) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                                                         |
|-------|----------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                              |
| 15-0  | GCHID[n] | 0     | Global channel interrupt disable bit. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.<br>Read: The corresponding channel is disabled for interrupt.<br>Write: No effect. |
|       |          | 1     | Read: The corresponding channel is enabled for interrupt.<br>Write: The corresponding channel is disabled for interrupt.                                                                            |

### 16.3.1.12 DMA Request Assignment Register 0 (DREQASI0)

**Figure 16-29. DMA Request Assignment Register 0 (DREQASI0) [offset = 54h]**

|          |    |         |    |          |    |         |    |
|----------|----|---------|----|----------|----|---------|----|
| 31       | 30 | 29      | 24 | 23       | 22 | 21      | 16 |
| Reserved |    | CH0ASI  |    | Reserved |    | CH1ASI  |    |
| R-0      |    | R/WP-0  |    | R-0      |    | R/WP-1h |    |
| 15       | 14 | 13      | 8  | 7        | 6  | 5       | 0  |
| Reserved |    | CH2ASI  |    | Reserved |    | CH3ASI  |    |
| R-0      |    | R/WP-2h |    | R-0      |    | R/WP-3h |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-21. DMA Request Assignment Register 0 (DREQASI0) Field Descriptions**

| Bit   | Field    | Value                    | Description                                                                                                                                                                                  |
|-------|----------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-30 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                       |
| 29-24 | CH0ASI   | 0<br>:<br>1Fh<br>20h-3Fh | Channel 0 assignment. This bit field chooses the DMA request assignment for channel 0.<br>DMA request line 0 triggers channel 0.<br>:<br>DMA request line 31 triggers channel 0.<br>Reserved |
| 23-22 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                       |
| 21-16 | CH1ASI   | 0<br>:<br>1Fh<br>20h-3Fh | Channel 1 assignment. This bit field chooses the DMA request assignment for channel 1.<br>DMA request line 0 triggers channel 1.<br>:<br>DMA request line 31 triggers channel 1.<br>Reserved |
| 15-14 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                       |
| 13-8  | CH2ASI   | 0<br>:<br>1Fh<br>20h-3Fh | Channel 2 assignment. This bit field chooses the DMA request assignment for channel 2.<br>DMA request line 0 triggers channel 2.<br>:<br>DMA request line 31 triggers channel 2.<br>Reserved |
| 7-6   | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                       |
| 5-0   | CH3ASI   | 0<br>:<br>1Fh<br>20h-3Fh | Channel 3 assignment. This bit field chooses the DMA request assignment for channel 3.<br>DMA request line 0 triggers channel 3.<br>:<br>DMA request line 31 triggers channel 3.<br>Reserved |

### 16.3.1.13 DMA Request Assignment Register 1 (DREQASI1)

**Figure 16-30. DMA Request Assignment Register 1 (DREQASI1) [offset = 58h]**

|          |    |         |    |          |    |         |    |
|----------|----|---------|----|----------|----|---------|----|
| 31       | 30 | 29      | 24 | 23       | 22 | 21      | 16 |
| Reserved |    | CH4ASI  |    | Reserved |    | CH5ASI  |    |
| R-0      |    | R/WP-4h |    | R-0      |    | R/WP-5h |    |
| 15       | 14 | 13      | 8  | 7        | 6  | 5       | 0  |
| Reserved |    | CH6ASI  |    | Reserved |    | CH7ASI  |    |
| R-0      |    | R/WP-6h |    | R-0      |    | R/WP-7h |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-22. DMA Request Assignment Register 1 (DREQASI1) Field Descriptions**

| Bit   | Field    | Value                    | Description                                                                                                                                                                                  |
|-------|----------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-30 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                       |
| 29-24 | CH4ASI   | 0<br>:<br>1Fh<br>20h-3Fh | Channel 4 assignment. This bit field chooses the DMA request assignment for channel 4.<br>DMA request line 0 triggers channel 4.<br>:<br>DMA request line 31 triggers channel 4.<br>Reserved |
| 23-22 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                       |
| 21-16 | CH5ASI   | 0<br>:<br>1Fh<br>20h-3Fh | Channel 5 assignment. This bit field chooses the DMA request assignment for channel 5.<br>DMA request line 0 triggers channel 5.<br>:<br>DMA request line 31 triggers channel 5.<br>Reserved |
| 15-14 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                       |
| 13-8  | CH6ASI   | 0<br>:<br>1Fh<br>20h-3Fh | Channel 6 assignment. This bit field chooses the DMA request assignment for channel 6.<br>DMA request line 0 triggers channel 6.<br>:<br>DMA request line 31 triggers channel 6.<br>Reserved |
| 7-6   | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                       |
| 5-0   | CH7ASI   | 0<br>:<br>1Fh<br>20h-3Fh | Channel 7 assignment. This bit field chooses the DMA request assignment for channel 7.<br>DMA request line 0 triggers channel 7.<br>:<br>DMA request line 31 triggers channel 7.<br>Reserved |

### 16.3.1.14 DMA Request Assignment Register 2 (DREQASI2)

**Figure 16-31. DMA Request Assignment Register 2 (DREQASI2) [offset = 5Ch]**

|          |    |         |    |          |    |         |    |
|----------|----|---------|----|----------|----|---------|----|
| 31       | 30 | 29      | 24 | 23       | 22 | 21      | 16 |
| Reserved |    | CH8ASI  |    | Reserved |    | CH9ASI  |    |
| R-0      |    | R/WP-8h |    | R-0      |    | R/WP-9h |    |
| 15       | 14 | 13      | 8  | 7        | 6  | 5       | 0  |
| Reserved |    | CH10ASI |    | Reserved |    | CH11ASI |    |
| R-0      |    | R/WP-Ah |    | R-0      |    | R/WP-Bh |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-23. DMA Request Assignment Register 2 (DREQASI2) Field Descriptions**

| Bit   | Field    | Value                    | Description                                                                                                                                                                                      |
|-------|----------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-30 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                           |
| 29-24 | CH8ASI   | 0<br>:<br>1Fh<br>20h-3Fh | Channel 8 assignment. This bit field chooses the DMA request assignment for channel 8.<br>DMA request line 0 triggers channel 8.<br>:<br>DMA request line 31 triggers channel 8.<br>Reserved     |
| 23-22 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                           |
| 21-16 | CH9ASI   | 0<br>:<br>1Fh<br>20h-3Fh | Channel 9 assignment. This bit field chooses the DMA request assignment for channel 9.<br>DMA request line 0 triggers channel 9.<br>:<br>DMA request line 31 triggers channel 9.<br>Reserved     |
| 15-14 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                           |
| 13-8  | CH10ASI  | 0<br>:<br>1Fh<br>20h-3Fh | Channel 10 assignment. This bit field chooses the DMA request assignment for channel 10.<br>DMA request line 0 triggers channel 10.<br>:<br>DMA request line 31 triggers channel 10.<br>Reserved |
| 7-6   | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                           |
| 5-0   | CH11ASI  | 0<br>:<br>1Fh<br>20h-3Fh | Channel 11 assignment. This bit field chooses the DMA request assignment for channel 11.<br>DMA request line 0 triggers channel 11.<br>:<br>DMA request line 31 triggers channel 11.<br>Reserved |

### 16.3.1.15 DMA Request Assignment Register 3 (DREQASI3)

**Figure 16-32. DMA Request Assignment Register 3 (DREQASI3) [offset = 60h]**

|          |    |         |    |          |    |         |    |
|----------|----|---------|----|----------|----|---------|----|
| 31       | 30 | 29      | 24 | 23       | 22 | 21      | 16 |
| Reserved |    | CH12ASI |    | Reserved |    | CH13ASI |    |
| R-0      |    | R/WP-Ch |    | R-0      |    | R/WP-Dh |    |
| 15       | 14 | 13      | 8  | 7        | 6  | 5       | 0  |
| Reserved |    | CH14ASI |    | Reserved |    | CH15ASI |    |
| R-0      |    | R/WP-Eh |    | R-0      |    | R/WP-Fh |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-24. DMA Request Assignment Register 3 (DREQASI3) Field Descriptions**

| Bit   | Field    | Value                    | Description                                                                                                                                                                                      |
|-------|----------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-30 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                           |
| 29-24 | CH12ASI  | 0<br>:<br>1Fh<br>20h-3Fh | Channel 12 assignment. This bit field chooses the DMA request assignment for channel 12.<br>DMA request line 0 triggers channel 12.<br>:<br>DMA request line 31 triggers channel 12.<br>Reserved |
| 23-22 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                           |
| 21-16 | CH13ASI  | 0<br>:<br>1Fh<br>20h-3Fh | Channel 13 assignment. This bit field chooses the DMA request assignment for channel 13.<br>DMA request line 0 triggers channel 13.<br>:<br>DMA request line 31 triggers channel 13.<br>Reserved |
| 15-14 | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                           |
| 13-8  | CH14ASI  | 0<br>:<br>1Fh<br>20h-3Fh | Channel 14 assignment. This bit field chooses the DMA request assignment for channel 14.<br>DMA request line 0 triggers channel 14.<br>:<br>DMA request line 31 triggers channel 14.<br>Reserved |
| 7-6   | Reserved | 0                        | Reads return 0. Writes have no effect.                                                                                                                                                           |
| 5-0   | CH15ASI  | 0<br>:<br>1Fh<br>20h-3Fh | Channel 15 assignment. This bit field chooses the DMA request assignment for channel 15.<br>DMA request line 0 triggers channel 15.<br>:<br>DMA request line 31 triggers channel 15.<br>Reserved |

### 16.3.1.16 Port Assignment Register 0 (PAR0)

**Figure 16-33. Port Assignment Register 0 (PAR0) [offset = 94h]**

|      |        |      |        |      |        |      |        |      |        |      |        |
|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|
| 31   | 30     | 28   | 27     | 26   | 24     | 23   | 22     | 20   | 19     | 18   | 16     |
| Rsvd | CH0PA  | Rsvd | CH1PA  | Rsvd | CH2PA  | Rsvd | CH3PA  | Rsvd | CH4PA  | Rsvd | CH5PA  |
| R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 |
| 15   | 14     | 12   | 11     | 10   | 8      | 7    | 6      | 4    | 3      | 2    | 0      |
| Rsvd | CH4PA  | Rsvd | CH5PA  | Rsvd | CH6PA  | Rsvd | CH7PA  | Rsvd | CH8PA  | Rsvd | CH9PA  |
| R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-25. Port Assignment Register 0 (PAR0) Field Descriptions**

| Bit   | Field    | Value      | Description                                                                           |
|-------|----------|------------|---------------------------------------------------------------------------------------|
| 31    | Reserved | 0          | Reads return 0. Writes have no effect.                                                |
| 30-28 | CH0PA    | 1xx<br>0xx | These bit fields determine to which port channel 0 is assigned.<br>Port B<br>Reserved |
| 27    | Reserved | 0          | Reads return 0. Writes have no effect.                                                |
| 26-24 | CH1PA    | 1xx<br>0xx | These bit fields determine to which port channel 1 is assigned.<br>Port B<br>Reserved |
| 23    | Reserved | 0          | Reads return 0. Writes have no effect.                                                |
| 22-20 | CH2PA    | 1xx<br>0xx | These bit fields determine to which port channel 2 is assigned.<br>Port B<br>Reserved |
| 19    | Reserved | 0          | Reads return 0. Writes have no effect.                                                |
| 18-16 | CH3PA    | 1xx<br>0xx | These bit fields determine to which port channel 3 is assigned.<br>Port B<br>Reserved |
| 15    | Reserved | 0          | Reads return 0. Writes have no effect.                                                |
| 14-12 | CH4PA    | 1xx<br>0xx | These bit fields determine to which port channel 4 is assigned.<br>Port B<br>Reserved |
| 11    | Reserved | 0          | Reads return 0. Writes have no effect.                                                |
| 10-8  | CH5PA    | 1xx<br>0xx | These bit fields determine to which port channel 5 is assigned.<br>Port B<br>Reserved |
| 7     | Reserved | 0          | Reads return 0. Writes have no effect.                                                |
| 6-4   | CH6PA    | 1xx<br>0xx | These bit fields determine to which port channel 6 is assigned.<br>Port B<br>Reserved |
| 3     | Reserved | 0          | Reads return 0. Writes have no effect.                                                |
| 2-0   | CH7PA    | 1xx<br>0xx | These bit fields determine to which port channel 7 is assigned.<br>Port B<br>Reserved |

### 16.3.1.17 Port Assignment Register 1 (PAR1)

**Figure 16-34. Port Assignment Register 1 (PAR1) [offset = 98h]**

|      |        |      |        |      |        |      |        |      |        |      |        |
|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|
| 31   | 30     | 28   | 27     | 26   | 24     | 23   | 22     | 20   | 19     | 18   | 16     |
| Rsvd | CH8PA  | Rsvd | CH9PA  | Rsvd | CH10PA | Rsvd | CH11PA | Rsvd | CH12PA | Rsvd | CH13PA |
| R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 |
| 15   | 14     | 12   | 11     | 10   | 8      | 7    | 6      | 4    | 3      | 2    | 0      |
| Rsvd | CH12PA | Rsvd | CH13PA | Rsvd | CH14PA | Rsvd | CH15PA | Rsvd | CH16PA | Rsvd | CH17PA |
| R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 | R-0  | R/WP-0 |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

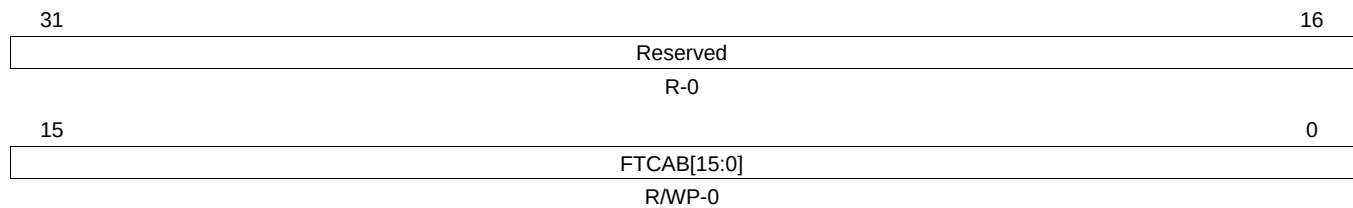
**Table 16-26. Port Assignment Register 1 (PAR1) Field Descriptions**

| Bit   | Field    | Value      | Description                                                                            |
|-------|----------|------------|----------------------------------------------------------------------------------------|
| 31    | Reserved | 0          | Reads return 0. Writes have no effect.                                                 |
| 30-28 | CH8PA    | 1xx<br>0xx | These bit fields determine to which port channel 8 is assigned.<br>Port B<br>Reserved  |
| 27    | Reserved | 0          | Reads return 0. Writes have no effect.                                                 |
| 26-24 | CH9PA    | 1xx<br>0xx | These bit fields determine to which port channel 9 is assigned.<br>Port B<br>Reserved  |
| 23    | Reserved | 0          | Reads return 0. Writes have no effect.                                                 |
| 22-20 | CH10PA   | 1xx<br>0xx | These bit fields determine to which port channel 10 is assigned.<br>Port B<br>Reserved |
| 19    | Reserved | 0          | Reads return 0. Writes have no effect.                                                 |
| 18-16 | CH11PA   | 1xx<br>0xx | These bit fields determine to which port channel 11 is assigned.<br>Port B<br>Reserved |
| 15    | Reserved | 0          | Reads return 0. Writes have no effect.                                                 |
| 14-12 | CH12PA   | 1xx<br>0xx | These bit fields determine to which port channel 12 is assigned.<br>Port B<br>Reserved |
| 11    | Reserved | 0          | Reads return 0. Writes have no effect.                                                 |
| 10-8  | CH13PA   | 1xx<br>0xx | These bit fields determine to which port channel 13 is assigned.<br>Port B<br>Reserved |
| 7     | Reserved | 0          | Reads return 0. Writes have no effect.                                                 |
| 6-4   | CH14PA   | 1xx<br>0xx | These bit fields determine to which port channel 14 is assigned.<br>Port B<br>Reserved |
| 3     | Reserved | 0          | Reads return 0. Writes have no effect.                                                 |
| 2-0   | CH15PA   | 1xx<br>0xx | These bit fields determine to which port channel 15 is assigned.<br>Port B<br>Reserved |

### 16.3.1.18 FTC Interrupt Mapping Register (FTCMAP)

**NOTE:** On this device Group B interrupts are not implemented; hence, user software should configure only Group A interrupts.

**Figure 16-35. FTC Interrupt Mapping Register (FTCMAP) [offset = B4h]**



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

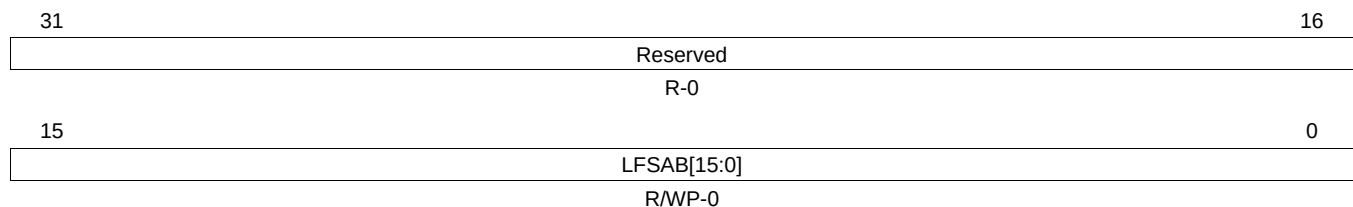
**Table 16-27. FTC Interrupt Mapping Register (FTCMAP) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                               |
|-------|----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                    |
| 15-0  | FTCAB[n] | 0     | Frame transfer complete (FTC) interrupt to Group A or Group B. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. |
|       |          | 0     | The FTC interrupt of the corresponding channel is routed to Group A.                                                                      |
|       |          | 1     | The FTC interrupt of the corresponding channel is routed to Group B.                                                                      |

### 16.3.1.19 LFS Interrupt Mapping Register (LFSMAP)

**NOTE:** On this device Group B interrupts are not implemented; hence, user software should configure only Group A interrupts.

**Figure 16-36. LFS Interrupt Mapping Register (LFSMAP) [offset = BCh]**



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-28. LFS Interrupt Mapping Register (LFSMAP) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                          |
|-------|----------|-------|--------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                               |
| 15-0  | LFSAB[n] | 0     | Last frame started (LFS) interrupt to Group A or Group B. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. |
|       |          | 0     | The LFS interrupt of the corresponding channel is routed to Group A.                                                                 |
|       |          | 1     | The LFS interrupt of the corresponding channel is routed to Group B.                                                                 |

### 16.3.1.20 HBC Interrupt Mapping Register (HBCMAP)

**NOTE:** On this device Group B interrupts are not implemented; hence, user software should configure only Group A interrupts.

**Figure 16-37. HBC Interrupt Mapping Register (HBCMAP) [offset = C4h]**

|        |             |    |
|--------|-------------|----|
| 31     | Reserved    | 16 |
| R-0    |             |    |
| 15     | HBCAB[15:0] | 0  |
| R/WP-0 |             |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-29. HBC Interrupt Mapping Register (HBCMAP) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                           |
|-------|----------|-------|---------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                |
| 15-0  | HBCAB[n] | 0     | Half block complete (HBC) interrupt to Group A or Group B. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. |
|       |          | 1     | The HBC interrupt of the corresponding channel is routed to Group A.                                                                  |
|       |          | 1     | The HBC interrupt of the corresponding channel is routed to Group B.                                                                  |

### 16.3.1.21 BTC Interrupt Mapping Register (BTCMAP)

**NOTE:** On this device Group B interrupts are not implemented; hence, user software should configure only Group A interrupts.

**Figure 16-38. BTC Interrupt Mapping Register (BTCMAP) [offset = CCh]**

|        |             |    |
|--------|-------------|----|
| 31     | Reserved    | 16 |
| R-0    |             |    |
| 15     | BTCAB[15:0] | 0  |
| R/WP-0 |             |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-30. BTC Interrupt Mapping Register (BTCMAP) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                               |
|-------|----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                    |
| 15-0  | BTCAB[n] | 0     | Block transfer complete (BTC) interrupt to Group A or Group B. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. |
|       |          | 1     | The BTC interrupt of the corresponding channel is routed to Group A.                                                                      |
|       |          | 1     | The BTC interrupt of the corresponding channel is routed to Group B.                                                                      |

### 16.3.1.22 FTC Interrupt Enable Set (FTCINTENAS)

**Figure 16-39. FTC Interrupt Enable Set (FTCINTENAS) [offset = DCh]**

|        |                 |    |
|--------|-----------------|----|
| 31     | Reserved        | 16 |
| R-0    |                 |    |
| 15     | FTCINTENA[15:0] | 0  |
| R/WP-0 |                 |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-31. FTC Interrupt Enable Set (FTCINTENAS) Field Descriptions**

| Bit   | Field        | Value | Description                                                                                                                |
|-------|--------------|-------|----------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved     | 0     | Reads return 0. Writes have no effect.                                                                                     |
| 15-0  | FTCINTENA[n] |       | Frame transfer complete (FTC) interrupt enable. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. |
|       |              | 0     | Read: The corresponding FTC interrupt of a channel is disabled.<br>Write: No effect.                                       |
|       |              | 1     | Read or write: The FTC interrupt of the corresponding channel is enabled.                                                  |

### 16.3.1.23 FTC Interrupt Enable Reset (FTCINTENAR)

**NOTE:** On this device Group B interrupts are not implemented hence user software should configure only Group A interrupts.

**Figure 16-40. FTC Interrupt Enable Reset (FTCINTENAR) [offset = E4h]**

|        |                 |    |
|--------|-----------------|----|
| 31     | Reserved        | 16 |
| R-0    |                 |    |
| 15     | FTCINTDIS[15:0] | 0  |
| R/WP-0 |                 |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-32. FTC Interrupt Enable Reset (FTCINTENAR) Field Descriptions**

| Bit   | Field        | Value | Description                                                                                                                 |
|-------|--------------|-------|-----------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved     | 0     | Reads return 0. Writes have no effect.                                                                                      |
| 15-0  | FTCINTDIS[n] |       | Frame transfer complete (FTC) interrupt disable. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. |
|       |              | 0     | Read: The corresponding FTC interrupt of a channel is disabled.<br>Write: No effect.                                        |
|       |              | 1     | Read: The corresponding FTC interrupt of a channel is enabled.<br>Write: The corresponding FTC interrupt is disabled.       |

### 16.3.1.24 LFS Interrupt Enable Set (LFSINTENAS)

**Figure 16-41. LFS Interrupt Enable Set (LFSINTENAS) [offset = ECh]**

|                 |    |
|-----------------|----|
| 31              | 16 |
| Reserved        |    |
| R-0             |    |
| 15              | 0  |
| LFSINTENA[15:0] |    |
| R/WP-0          |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-33. LFS Interrupt Enable Set (LFSINTENAS) Field Descriptions**

| Bit   | Field        | Value | Description                                                                                                           |
|-------|--------------|-------|-----------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved     | 0     | Reads return 0. Writes have no effect.                                                                                |
| 15-0  | LFSINTENA[n] |       | Last frame started (LFS) interrupt enable. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. |
|       |              | 0     | Read: The corresponding LFS interrupt of a channel is disabled.<br>Write: No effect.                                  |
|       |              | 1     | Read or write: The LFS interrupt of the corresponding channel is enabled.                                             |

### 16.3.1.25 LFS Interrupt Enable Reset (LFSINTENAR)

**Figure 16-42. LFS Interrupt Enable Reset (LFSINTENAR) [offset = F4h]**

|                 |    |
|-----------------|----|
| 31              | 16 |
| Reserved        |    |
| R-0             |    |
| 15              | 0  |
| LFSINTDIS[15:0] |    |
| R/WP-0          |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-34. LFS Interrupt Enable Reset (LFSINTENAR) Field Descriptions**

| Bit   | Field        | Value | Description                                                                                                                            |
|-------|--------------|-------|----------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved     | 0     | Reads return 0. Writes have no effect.                                                                                                 |
| 15-0  | LFSINTDIS[n] |       | Last frame started (LFS) interrupt disable. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.                 |
|       |              | 0     | Read: The LFS interrupt of the corresponding channel is disabled.<br>Write: No effect.                                                 |
|       |              | 1     | Read: The LFS interrupt of the corresponding channel is enabled.<br>Write: The LFS interrupt of the corresponding channel is disabled. |

### 16.3.1.26 HBC Interrupt Enable Reset (HBCINTENAS)

**Figure 16-43. HBC Interrupt Enable Set (HBCINTENAS) [offset = FCh]**

|                 |    |
|-----------------|----|
| 31              | 16 |
| Reserved        |    |
| R-0             |    |
| 15              | 0  |
| HBCINTENA[15:0] |    |
| R/WP-0          |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-35. HBC Interrupt Enable Set (HBCINTENAS) Field Descriptions**

| Bit   | Field        | Value | Description                                                                                                            |
|-------|--------------|-------|------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved     | 0     | Reads return 0. Writes have no effect.                                                                                 |
| 15-0  | HBCINTENA[n] |       | Half block complete (HBC) interrupt enable. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. |
|       |              | 0     | Read: The HBC interrupt of the corresponding channel is disabled.<br>Write: No effect.                                 |
|       |              | 1     | Read or write: The HBC interrupt of the corresponding channel is enabled.                                              |

### 16.3.1.27 HBC Interrupt Enable Reset (HBCINTENAR)

**Figure 16-44. HBC Interrupt Enable Reset (HBCINTENAR) [offset = 104h]**

|                 |    |
|-----------------|----|
| 31              | 16 |
| Reserved        |    |
| R-0             |    |
| 15              | 0  |
| HBCINTDIS[15:0] |    |
| R/WP-0          |    |

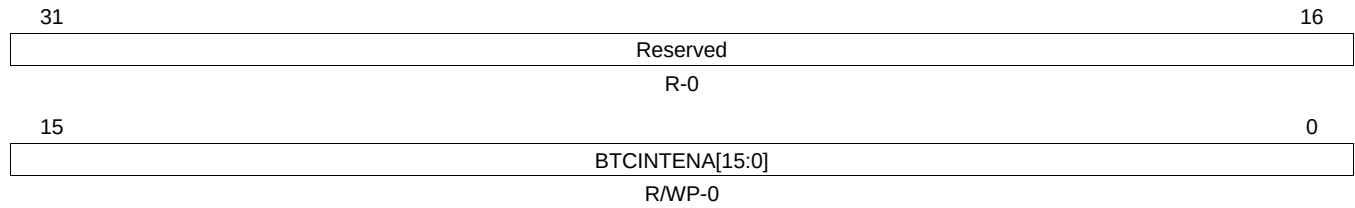
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-36. HBC Interrupt Enable Reset (HBCINTENAR) Field Descriptions**

| Bit   | Field        | Value | Description                                                                                                                            |
|-------|--------------|-------|----------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved     | 0     | Reads return 0. Writes have no effect.                                                                                                 |
| 15-0  | HBCINTDIS[n] |       | Half block complete (HBC) interrupt disable. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.                |
|       |              | 0     | Read: The HBC interrupt of the corresponding channel is disabled.<br>Write: No effect.                                                 |
|       |              | 1     | Read: The HBC interrupt of the corresponding channel is enabled.<br>Write: The HBC interrupt of the corresponding channel is disabled. |

### 16.3.1.28 BTC Interrupt Enable Set (BTCINTENAS)

**Figure 16-45. BTC Interrupt Enable Set (BTCINTENAS) [offset = 10Ch]**



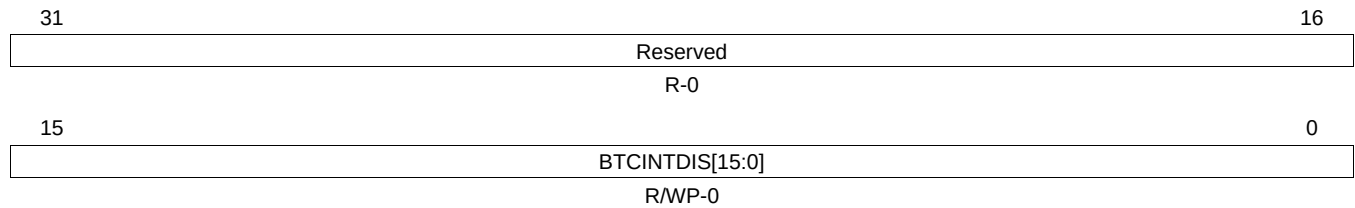
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-37. BTC Interrupt Enable Reset (BTCINTENAS) Field Descriptions**

| Bit   | Field        | Value | Description                                                                                                                |
|-------|--------------|-------|----------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved     | 0     | Reads return 0. Writes have no effect.                                                                                     |
| 15-0  | BTCINTENA[n] |       | Block transfer complete (BTC) interrupt enable. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. |
|       |              | 0     | Read: The BTC interrupt of the corresponding channel is disabled.<br>Write: No effect.                                     |
|       |              | 1     | Read or write: The BTC interrupt of the corresponding channel is enabled.                                                  |

### 16.3.1.29 BTC Interrupt Enable Reset (BTCINTENAR)

**Figure 16-46. BTC Interrupt Enable Reset (BTCINTENAR) [offset = 114h]**



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-38. BTC Interrupt Enable Reset (BTCINTENAR) Field Descriptions**

| Bit   | Field        | Value | Description                                                                                                                            |
|-------|--------------|-------|----------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved     | 0     | Reads return 0. Writes have no effect.                                                                                                 |
| 15-0  | BTCINTDIS[n] |       | Block transfer complete (BTC) interrupt disable. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.            |
|       |              | 0     | Read: The BTC interrupt of the corresponding channel is disabled.<br>Write: No effect.                                                 |
|       |              | 1     | Read: The BTC interrupt of the corresponding channel is enabled.<br>Write: The BTC interrupt of the corresponding channel is disabled. |

### 16.3.1.30 Global Interrupt Flag Register (GINTFLAG)

**Figure 16-47. Global Interrupt Flag Register (GINTFLAG) [offset = 11Ch]**

|    |            |    |
|----|------------|----|
| 31 | Reserved   | 16 |
|    | R-0        |    |
| 15 | GINT[15:0] | 0  |
|    | R/WP-0     |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-39. Global Interrupt Flag Register (GINTFLAG) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                                                                                                          |
|-------|----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                               |
| 15-0  | GINT[n]  | 0     | Global interrupt flags. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on. A global interrupt flag bit is an OR function of FTC, LFS, HBC, and BTC interrupt flags.<br>No interrupt is pending on the corresponding channel. |
|       |          | 1     | One or more of the interrupt types (FTC, LFS, HBC, or BTC) is pending on the corresponding channel.                                                                                                                                                  |

### 16.3.1.31 FTC Interrupt Flag Register (FTCFLAG)

**Figure 16-48. FTC Interrupt Flag Register (FTCFLAG) [offset = 124h]**

|    |            |    |
|----|------------|----|
| 31 | Reserved   | 16 |
|    | R-0        |    |
| 15 | FTCI[15:0] | 0  |
|    | R/W1CP-0   |    |

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode to clear the bit; -n = value after reset

**Table 16-40. FTC Interrupt Flag Register (FTCFLAG) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|----------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 15-0  | FTCI[n]  | 0     | Frame transfer complete (FTC) flags. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.<br><b>Note: Reading from the respective interrupt channel offset register also clears the corresponding flag (see <a href="#">Section 16.3.1.36</a> and <a href="#">Section 16.3.1.40</a>).</b><br><b>Note: The state of the flag bit can be polled even if the corresponding interrupt enable bit is cleared.</b><br>Read: An FTC interrupt of the corresponding channel is not pending.<br>Write: No effect. |
|       |          | 1     | Read: An FTC interrupt of the corresponding channel is pending.<br>Write: The flag is cleared.                                                                                                                                                                                                                                                                                                                                                                                                                                 |

### 16.3.1.32 LFS Interrupt Flag Register (LFSFLAG)

**Figure 16-49. LFS Interrupt Flag Register (LFSFLAG) [offset = 12Ch]**

|            |    |
|------------|----|
| 31         | 16 |
| Reserved   |    |
| R-0        |    |
| 15         | 0  |
| LFSI[15:0] |    |
| R/W1CP-0   |    |

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode to clear the bit; -n = value after reset

**Table 16-41. LFS Interrupt Flag Register (LFSFLAG) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|----------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 15-0  | LFSI[n]  | 0     | Last frame started (LFS) flags. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.<br><br><b>Note: Reading from the respective interrupt channel offset register also clears the corresponding flag (see <a href="#">Section 16.3.1.37</a> and <a href="#">Section 16.3.1.41</a>).</b><br><br><b>Note: The state of the flag bit can be polled even if the corresponding interrupt enable bit is cleared.</b><br>Read: An LFS interrupt of the corresponding channel is not pending.<br>Write: No effect. |
|       |          | 1     | Read: An LFS interrupt of the corresponding channel is pending.<br>Write: The flag is cleared.                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### 16.3.1.33 HBC Interrupt Flag Register (HBCFLAG)

**Figure 16-50. HBC Interrupt Flag Register (HBCFLAG) [offset = 134h]**

|            |    |
|------------|----|
| 31         | 16 |
| Reserved   |    |
| R-0        |    |
| 15         | 0  |
| HBCI[15:0] |    |
| R/W1CP-0   |    |

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode to clear the bit; -n = value after reset

**Table 16-42. HBC Interrupt Flag (HBCFLAG) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|----------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 15-0  | HBCI[n]  | 0     | Half block transfer (HBC) complete flags. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.<br><br><b>Note: Reading from the respective interrupt channel offset register also clears the corresponding flag (see <a href="#">Section 16.3.1.38</a> and <a href="#">Section 16.3.1.42</a>).</b><br><br><b>Note: The state of the flag bit can be polled even if the corresponding interrupt enable bit is cleared.</b><br>Read: An HBC interrupt of the corresponding channel is not pending.<br>Write: No effect. |
|       |          | 1     | Read: An HBC interrupt of the corresponding channel is pending.<br>Write: The flag is cleared.                                                                                                                                                                                                                                                                                                                                                                                                                                              |

### 16.3.1.34 BTC Interrupt Flag Register (BTCFLAG)

**Figure 16-51. BTC Interrupt Flag Register (BTCFLAG) [offset = 13Ch]**

|    |            |    |
|----|------------|----|
| 31 | Reserved   | 16 |
|    | R-0        |    |
| 15 | BTCI[15:0] | 0  |
|    | R/W1CP-0   |    |

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode to clear the bit; -n = value after reset

**Table 16-43. BTC Interrupt Flag Register (BTCFLAG) Field Descriptions**

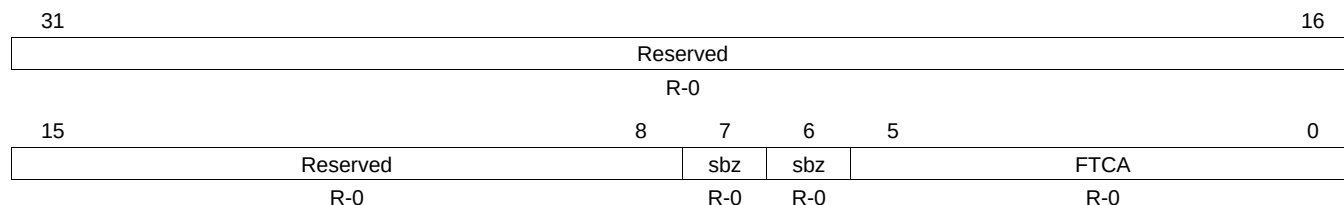
| Bit   | Field    | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|----------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 15-0  | BTCI[n]  | 0     | Block transfer complete (BTC) flags. Bit 0 corresponds to channel 0, bit 1 corresponds to channel 1, and so on.<br><br><b>Note:</b> Reading from the respective interrupt channel offset register also clears the corresponding flag (see <a href="#">Section 16.3.1.39</a> and <a href="#">Section 16.3.1.43</a> ).<br><br><b>Note:</b> The state of the flag bit can be polled even if the corresponding interrupt enable bit is cleared.<br><br>Read: A BTC interrupt of the corresponding channel is not pending.<br>Write: No effect. |
|       |          | 1     | Read: A BTC interrupt of the corresponding channel is pending.<br>Write: The flag is cleared.                                                                                                                                                                                                                                                                                                                                                                                                                                              |

### 16.3.1.35 BER Interrupt Flag Register (BERFLAG)

The BERFLAG will never be set in this device. The bus error reporting is handled by the DMA Read Imprecise Error and DMA Write Imprecise Error asserted to the ESM module directly, which are detected at the device level. See the ESM error mapping for the DMA Read/Write Imprecise Error.

### 16.3.1.36 FTCA Interrupt Channel Offset Register (FTCAOFFSET)

**Figure 16-52. FTCA Interrupt Channel Offset Register (FTCAOFFSET) [offset = 14Ch]**



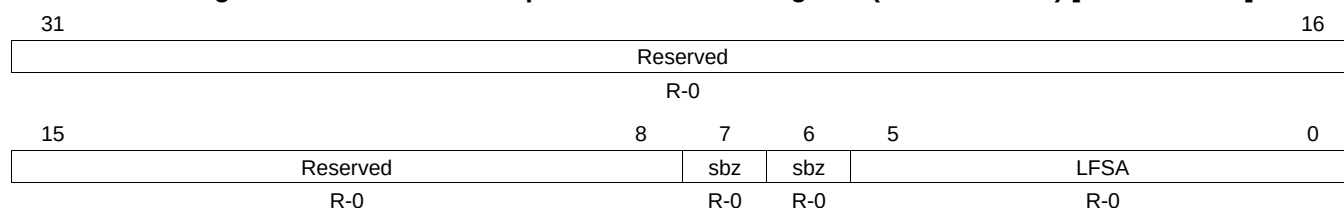
LEGEND: R = Read only; -n = value after reset

**Table 16-44. FTCA Interrupt Channel Offset Register (FTCAOFFSET) Field Descriptions**

| Bit  | Field    | Value                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|----------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-8 | Reserved | 0                                                                         | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 7-6  | sbz      | 0                                                                         | These bits should always be programmed as 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5-0  | FTCA     | <div>0</div> <div>1h</div> <div>:</div> <div>10h</div> <div>11h-3Fh</div> | <div>Channel causing FTC interrupt Group A. These bits contain the channel number of the pending interrupt for Group A if the corresponding interrupt enable is set.</div> <div><b>Note: Reading this location clears the corresponding interrupt pending flag (see <a href="#">Section 16.3.1.31</a>) with the highest priority.</b></div> <div>No interrupt is pending.</div> <div>Channel 0 is causing the pending interrupt Group A.</div> <div>:</div> <div>Channel 15 is causing the pending interrupt Group A.</div> <div>Reserved</div> |

### 16.3.1.37 LFSA Interrupt Channel Offset Register (LFSAOFFSET)

**Figure 16-53. LFSA Interrupt Channel Offset Register (LFSAOFFSET) [offset = 150h]**



LEGEND: R = Read only; -n = value after reset

**Table 16-45. LFSA Interrupt Channel Offset Register (LFSAOFFSET) Field Descriptions**

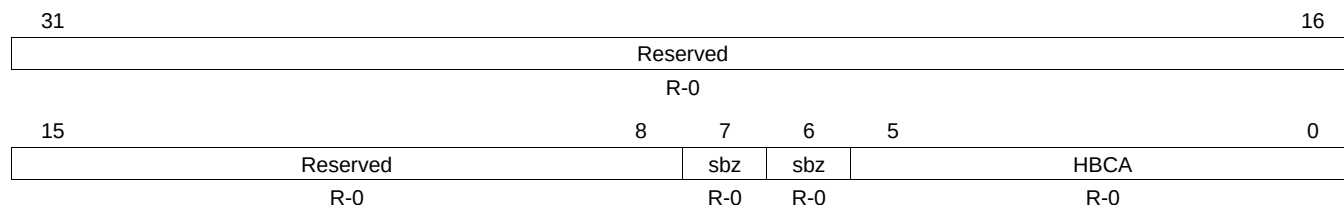
| Bit  | Field    | Value | Description                                  |
|------|----------|-------|----------------------------------------------|
| 31-8 | Reserved | 0     | Reads return 0. Writes have no effect.       |
| 7-6  | sbz      | 0     | These bits should always be programmed as 0. |

**Table 16-45. LFSA Interrupt Channel Offset Register (LFSAOFFSET) Field Descriptions (continued)**

| Bit | Field | Value   | Description                                                                                                                                                                                                                                                                                                              |
|-----|-------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5-0 | LFSA  |         | Channel causing LFS interrupt Group A. These bits contain the channel number of the pending interrupt for Group A if the corresponding interrupt enable is set.<br><b>Note: Reading this location clears the corresponding interrupt pending flag (see <a href="#">Section 16.3.1.32</a>) with the highest priority.</b> |
|     |       | 0       | No interrupt is pending.                                                                                                                                                                                                                                                                                                 |
|     |       | 1h      | Channel 0 is causing the pending interrupt Group A.                                                                                                                                                                                                                                                                      |
|     |       | :       | :                                                                                                                                                                                                                                                                                                                        |
|     |       | 10h     | Channel 15 is causing the pending interrupt Group A.                                                                                                                                                                                                                                                                     |
|     |       | 11h-3Fh | Reserved                                                                                                                                                                                                                                                                                                                 |

### 16.3.1.38 HBCA Interrupt Channel Offset Register (HBCAOFFSET)

**Figure 16-54. HBCA Interrupt Channel Offset Register (HBCAOFFSET) [offset = 154h]**



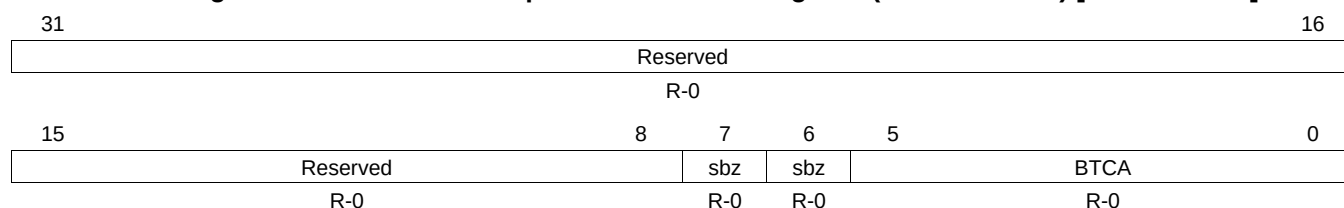
LEGEND: R = Read only; -n = value after reset

**Table 16-46. HBCA Interrupt Channel Offset Register (HBCAOFFSET) Field Descriptions**

| Bit  | Field    | Value   | Description                                                                                                                                                                                                                                                                                                              |
|------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-8 | Reserved | 0       | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                   |
| 7-6  | sbz      | 0       | These bits should always be programmed as 0.                                                                                                                                                                                                                                                                             |
| 5-0  | HBCA     |         | Channel causing HBC interrupt Group A. These bits contain the channel number of the pending interrupt for Group A if the corresponding interrupt enable is set.<br><b>Note: Reading this location clears the corresponding interrupt pending flag (see <a href="#">Section 16.3.1.33</a>) with the highest priority.</b> |
|      |          | 0       | No interrupt is pending.                                                                                                                                                                                                                                                                                                 |
|      |          | 1h      | Channel 0 is causing the pending interrupt Group A.                                                                                                                                                                                                                                                                      |
|      |          | :       | :                                                                                                                                                                                                                                                                                                                        |
|      |          | 10h     | Channel 15 is causing the pending interrupt Group A.                                                                                                                                                                                                                                                                     |
|      |          | 11h-3Fh | Reserved                                                                                                                                                                                                                                                                                                                 |

### 16.3.1.39 BTCA Interrupt Channel Offset Register (BTCAOFFSET)

**Figure 16-55. BTCA Interrupt Channel Offset Register (BTCAOFFSET) [offset = 158h]**



LEGEND: R = Read only; -n = value after reset

**Table 16-47. BTCA Interrupt Channel Offset Register (BTCAOFFSET) Field Descriptions**

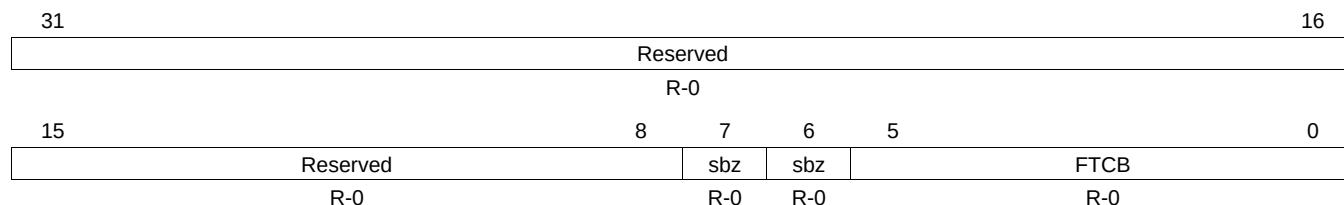
| Bit  | Field    | Value | Description                                  |
|------|----------|-------|----------------------------------------------|
| 31-8 | Reserved | 0     | Reads return 0. Writes have no effect.       |
| 7-6  | sbz      | 0     | These bits should always be programmed as 0. |

**Table 16-47. BTCA Interrupt Channel Offset Register (BTCAOFFSET) Field Descriptions (continued)**

| Bit | Field | Value   | Description                                                                                                                                                                                                                                                                                                              |
|-----|-------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5-0 | BTCA  |         | Channel causing BTC interrupt Group A. These bits contain the channel number of the pending interrupt for Group A if the corresponding interrupt enable is set.<br><b>Note: Reading this location clears the corresponding interrupt pending flag (see <a href="#">Section 16.3.1.34</a>) with the highest priority.</b> |
|     |       | 0       | No interrupt is pending.                                                                                                                                                                                                                                                                                                 |
|     |       | 1h      | Channel 0 is causing the pending interrupt Group A.                                                                                                                                                                                                                                                                      |
|     |       | :       | :                                                                                                                                                                                                                                                                                                                        |
|     |       | 10h     | Channel 15 is causing the pending interrupt Group A.                                                                                                                                                                                                                                                                     |
|     |       | 11h-3Fh | Reserved                                                                                                                                                                                                                                                                                                                 |

### 16.3.1.40 FTCB Interrupt Channel Offset Register (FTCBOFFSET)

**Figure 16-56. FTCB Interrupt Channel Offset Register (FTCBOFFSET) [offset = 160h]**



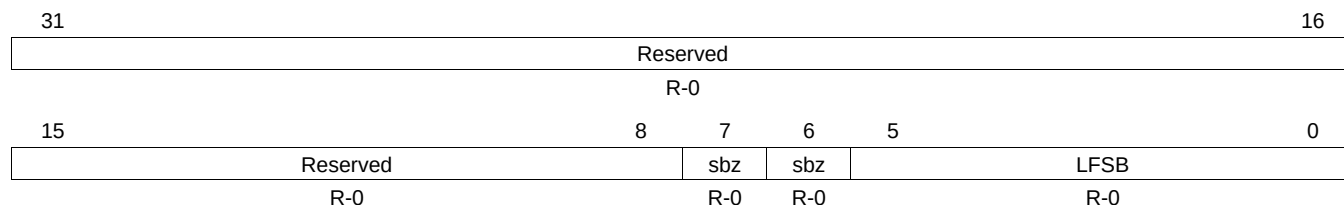
LEGEND: R = Read only; -n = value after reset

**Table 16-48. FTCB Interrupt Channel Offset Register (FTCBOFFSET) Field Descriptions**

| Bit  | Field    | Value   | Description                                                                                                                                                                                                                                                                                                              |
|------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-8 | Reserved | 0       | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                   |
| 7-6  | sbz      | 0       | These bits should always be programmed as 0.                                                                                                                                                                                                                                                                             |
| 5-0  | FTCB     |         | Channel causing FTC interrupt Group B. These bits contain the channel number of the pending interrupt for Group B if the corresponding interrupt enable is set.<br><b>Note: Reading this location clears the corresponding interrupt pending flag (see <a href="#">Section 16.3.1.31</a>) with the highest priority.</b> |
|      |          | 0       | No interrupt is pending.                                                                                                                                                                                                                                                                                                 |
|      |          | 1h      | Channel 0 is causing the pending interrupt Group B.                                                                                                                                                                                                                                                                      |
|      |          | :       | :                                                                                                                                                                                                                                                                                                                        |
|      |          | 10h     | Channel 15 is causing the pending interrupt Group B.                                                                                                                                                                                                                                                                     |
|      |          | 11h-3Fh | Reserved                                                                                                                                                                                                                                                                                                                 |

### 16.3.1.41 LFSB Interrupt Channel Offset Register (LFSBOFFSET)

**Figure 16-57. LFSB Interrupt Channel Offset Register (LFSBOFFSET) [offset = 164h]**



LEGEND: R = Read only; -n = value after reset

**Table 16-49. LFSB Interrupt Channel Offset Register (LFSBOFFSET) Field Descriptions**

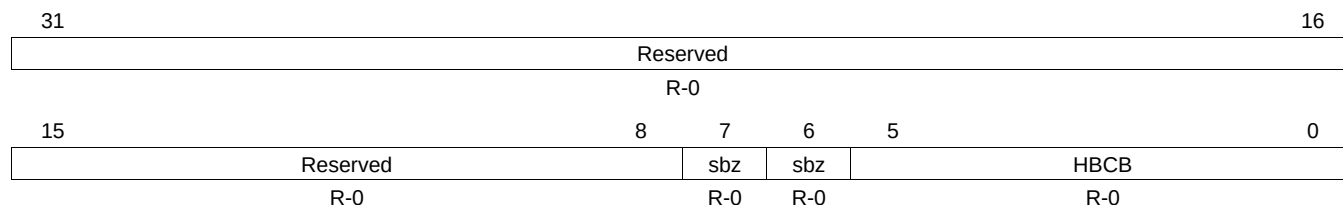
| Bit  | Field    | Value | Description                                  |
|------|----------|-------|----------------------------------------------|
| 31-8 | Reserved | 0     | Reads return 0. Writes have no effect.       |
| 7-6  | sbz      | 0     | These bits should always be programmed as 0. |

**Table 16-49. LFSB Interrupt Channel Offset Register (LFSBOFFSET) Field Descriptions (continued)**

| Bit | Field | Value   | Description                                                                                                                                                                                                                                                                                                              |
|-----|-------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5-0 | LFSB  |         | Channel causing LFS interrupt Group B. These bits contain the channel number of the pending interrupt for Group B if the corresponding interrupt enable is set.<br><b>Note: Reading this location clears the corresponding interrupt pending flag (see <a href="#">Section 16.3.1.32</a>) with the highest priority.</b> |
|     |       | 0       | No interrupt is pending.                                                                                                                                                                                                                                                                                                 |
|     |       | 1h      | Channel 0 is causing the pending interrupt Group B.                                                                                                                                                                                                                                                                      |
|     |       | :       | :                                                                                                                                                                                                                                                                                                                        |
|     |       | 10h     | Channel 15 is causing the pending interrupt Group B.                                                                                                                                                                                                                                                                     |
|     |       | 11h-3Fh | Reserved                                                                                                                                                                                                                                                                                                                 |

### 16.3.1.42 HBCB Interrupt Channel Offset Register (HBCBOFFSET)

**Figure 16-58. HBCB Interrupt Channel Offset Register (HBCBOFFSET) [offset = 168h]**



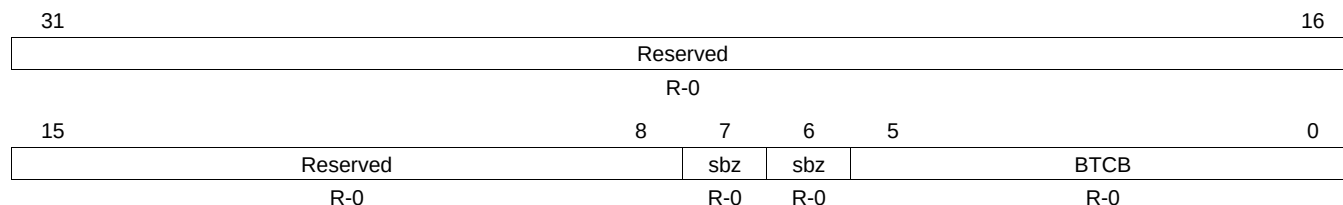
LEGEND: R = Read only; -n = value after reset

**Table 16-50. HBCB Interrupt Channel Offset Register (HBCBOFFSET) Field Descriptions**

| Bit  | Field    | Value                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|----------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-8 | Reserved | 0                                                                         | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 7-6  | sbz      | 0                                                                         | These bits should always be programmed as 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5-0  | HBCB     | <div>0</div> <div>1h</div> <div>:</div> <div>10h</div> <div>11h-3Fh</div> | <div>Channel causing HBC interrupt Group B. These bits contain the channel number of the pending interrupt for Group B if the corresponding interrupt enable is set.</div> <div><b>Note: Reading this location clears the corresponding interrupt pending flag (see <a href="#">Section 16.3.1.33</a>) with the highest priority.</b></div> <div>No interrupt is pending.</div> <div>Channel 0 is causing the pending interrupt Group B.</div> <div>:</div> <div>Channel 15 is causing the pending interrupt Group B.</div> <div>Reserved</div> |

### 16.3.1.43 BTCB Interrupt Channel Offset Register (BTCBOFFSET)

**Figure 16-59. BTCB Interrupt Channel Offset Register (BTCBOFFSET) [offset = 16Ch]**



LEGEND: R = Read only; -n = value after reset

**Table 16-51. BTCB Interrupt Channel Offset Register (BTCBOFFSET) Field Descriptions**

| Bit  | Field    | Value | Description                                  |
|------|----------|-------|----------------------------------------------|
| 31-8 | Reserved | 0     | Reads return 0. Writes have no effect.       |
| 7-6  | sbz      | 0     | These bits should always be programmed as 0. |

**Table 16-51. BTCB Interrupt Channel Offset Register (BTCBOFFSET) Field Descriptions (continued)**

| Bit | Field | Value   | Description                                                                                                                                                                                                                                                                                                                  |
|-----|-------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5-0 | BTCB  |         | Channel causing BTC interrupt Group B. These bits contain the channel number of the pending interrupt for Group B if the corresponding interrupt enable is set.<br><br><b>Note: Reading this location clears the corresponding interrupt pending flag (see <a href="#">Section 16.3.1.34</a>) with the highest priority.</b> |
|     |       | 0       | No interrupt is pending.                                                                                                                                                                                                                                                                                                     |
|     |       | 1h      | Channel 0 is causing the pending interrupt Group B.                                                                                                                                                                                                                                                                          |
|     |       | :       | :                                                                                                                                                                                                                                                                                                                            |
|     |       | 10h     | Channel 15 is causing the pending interrupt Group B.                                                                                                                                                                                                                                                                         |
|     |       | 11h-3Fh | Reserved                                                                                                                                                                                                                                                                                                                     |

### 16.3.1.44 Port Control Register (PTCRL)

**Figure 16-60. Port Control Register (PTCRL) [offset = 178h]**

|          |  |  |  |  |  |  |  |  |  |  |        |         |         |    |  |    |    |       |
|----------|--|--|--|--|--|--|--|--|--|--|--------|---------|---------|----|--|----|----|-------|
| 31       |  |  |  |  |  |  |  |  |  |  |        |         |         |    |  | 25 | 24 |       |
| Reserved |  |  |  |  |  |  |  |  |  |  |        |         |         |    |  |    |    | PENDB |
| R-0      |  |  |  |  |  |  |  |  |  |  |        |         |         |    |  |    |    | R-0   |
| 23       |  |  |  |  |  |  |  |  |  |  | 19     | 18      | 17      | 16 |  |    |    |       |
| Reserved |  |  |  |  |  |  |  |  |  |  | BYB    | PSFRHQP | PSFRLQP |    |  |    |    |       |
| R-0      |  |  |  |  |  |  |  |  |  |  | R/WP-0 | R/WP-0  | R/WP-0  |    |  |    |    |       |
| 15       |  |  |  |  |  |  |  |  |  |  |        |         |         |    |  |    | 0  |       |
| Reserved |  |  |  |  |  |  |  |  |  |  |        |         |         |    |  |    |    |       |
| R-0      |  |  |  |  |  |  |  |  |  |  |        |         |         |    |  |    |    |       |

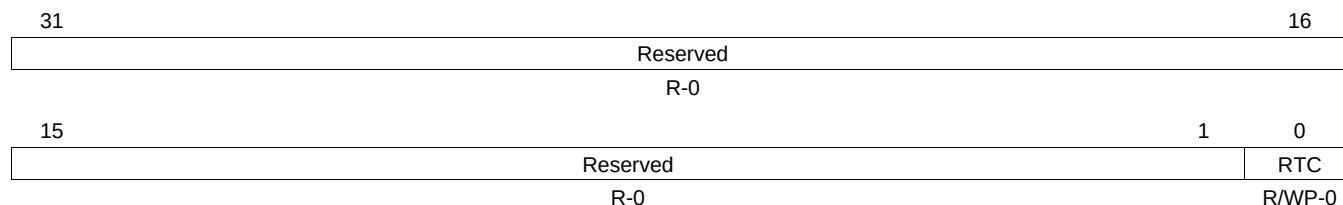
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-52. Port Control Register (PTCRL) Field Descriptions**

| Bit   | Field    | Value  | Description                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|----------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-25 | Reserved | 0      | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                        |
| 24    | PENDB    | 0<br>1 | Transfers pending for Port B. This flag determines if transfers are ongoing on port B. The flag will be cleared if no transfers are performed. It can be used to determine if there is still data transferred while DMA_EN is set to 0 in GCTCRL. In this case, once all transfers are finished, the flag will be set to 0.<br>No transfers are pending.<br>Transfers are pending.            |
| 23-19 | Reserved | 0      | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                        |
| 18    | BYB      | 0<br>1 | Bypass FIFO B.<br>FIFO B is not bypassed.<br>FIFO B is bypassed. Writing 1 to this bit limits the FIFO depth to the size of one element. That means that after one element is read, the write-out to the destination will begin. This feature is particularly useful to minimize switching latency between channels.<br><b>Note: This feature does not make optimal use of bus bandwidth.</b> |
| 17    | PSFRHQP  | 0<br>1 | Priority scheme fix or rotate for high priority queue of Port B.<br>Fixed priority is used.<br>Rotation priority is used.                                                                                                                                                                                                                                                                     |
| 16    | PSFRLQP  | 0<br>1 | Priority scheme fix or rotate for low priority queue of Port B.<br>The fixed priority scheme is used.<br>The rotation priority scheme is used.                                                                                                                                                                                                                                                |
| 15-0  | Reserved | 0      | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                        |

### 16.3.1.45 RAM Test Control (RTCTRL)

**Figure 16-61. RAM Test Control (RTCTRL) [offset = 17Ch]**



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-53. RAM Test Control (RTCTRL) Field Descriptions**

| Bit  | Field    | Value | Description                                                                                                                                                                                                             |
|------|----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-1 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                  |
| 0    | RTC      |       | RAM Test Control. Writing a 1 to this bit opens the write access to the reserved locations of control packet RAM as defined in the memory map.<br><b>Note: This bit should be cleared to 0 during normal operation.</b> |
|      |          | 0     | RAM Test Control is disabled.                                                                                                                                                                                           |
|      |          | 1     | RAM Test Control is enabled.                                                                                                                                                                                            |

### 16.3.1.46 Debug Control (DCTRL)

**Figure 16-62. Debug Control (DCTRL) [offset = 180h]**

|          |    |       |    |          |          |         |
|----------|----|-------|----|----------|----------|---------|
| 31       | 29 | 28    | 24 | 23       | 17       | 16      |
| Reserved |    | CHNUM |    | Reserved |          | DMADBGS |
| R-0      |    | R-0   |    | R-0      |          | R/W1C-0 |
| 15       |    |       |    |          | 1        | 0       |
|          |    |       |    |          | Reserved | DBGEN   |
|          |    |       |    |          | R-0      | R/WC-0  |

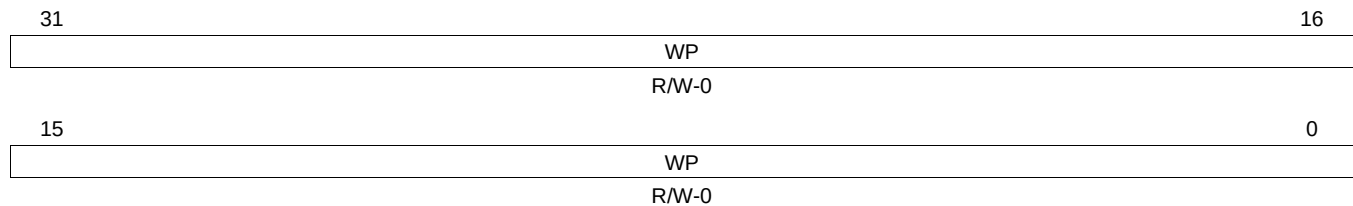
LEGEND: R/W = Read/Write; R = Read only; W1C = Write 1 to clear; -n = value after reset

**Table 16-54. Debug Control (DCTRL) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|----------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-29 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                     |
| 28-24 | CHNUM    | 0-1Fh | Channel Number. This bit field indicates the channel number that causes the watch point to match.                                                                                                                                                                                                                                                                                                                          |
| 23-17 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                     |
| 16    | DMADBGS  | 0     | DMA debug status. When a watch point is set up to watch for a unique bus address or a range of addresses is true on one of the three bus ports, then the DMA debug status bit is set to 1 and a debug request signal is asserted to the main CPU. The CPU must write a 1 to clear this bit for the DMA controller to release the debug request signal.<br>Read: No watch point condition is detected.<br>Write: No effect. |
|       |          | 1     | Read: The watch point condition is detected.<br>Write: The bit is cleared.                                                                                                                                                                                                                                                                                                                                                 |
| 15-1  | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                     |
| 0     | DBGEN    | 0     | Debug Enable.<br><b>Note: This bit can only be set when using a debugger.</b><br><b>Note: This bit is reset when Test reset (nTRST) is low.</b><br>Debug is disabled.                                                                                                                                                                                                                                                      |
|       |          | 1     | The watch point checking logics is enabled.                                                                                                                                                                                                                                                                                                                                                                                |

### 16.3.1.47 Watch Point Register (WPR)

**Figure 16-63. Watch Point Register (WPR) [offset = 184h]**



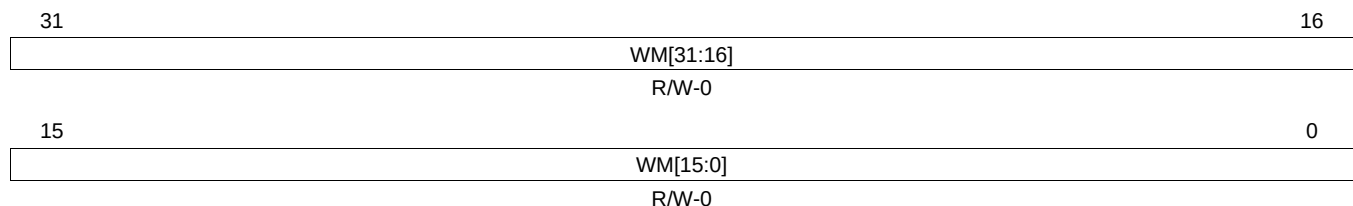
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

**Table 16-55. Watch Point Register (WPR) Field Descriptions**

| Bit  | Field | Value        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|-------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | WP    | 0-FFFF FFFFh | <p>Watch point.</p> <p><b>Note:</b> These bits can only be set when using a debugger.</p> <p><b>Note:</b> This register is only reset by a when Test reset (nTRST).</p> <p>A 32-bit address can be programmed into this register as a watch point. This register is used with the watch mask register (WMR).</p> <p>When the DBGEN bit in the DCTRL register is set and a unique address or a range of addresses are detected on the AHB address bus of Port B, a debug request signal is sent to the ARM CPU. The state machine of the port in which the watch point condition is true is frozen.</p> |

### 16.3.1.48 Watch Mask Register (WMR)

**Figure 16-64. Watch Mask Register (WMR) [offset = 188h]**



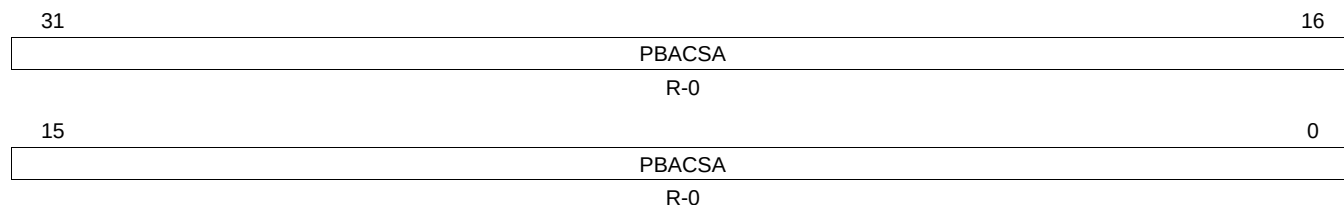
LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

**Table 16-56. Watch Mask Register (WMR) Field Descriptions**

| Bit  | Field | Value | Description                                                                                                                                                            |
|------|-------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | WM[n] |       | <p>Watch mask.</p> <p><b>Note:</b> These bits can only be set when using a debugger.</p> <p><b>Note:</b> This register is only reset by a when Test reset (nTRST).</p> |
|      |       | 0     | Allows the corresponding bit in the WPR register to be used for address matching for a watch point.                                                                    |
|      |       | 1     | Masks the corresponding bit in the WPR register and is disregarded in the comparison.                                                                                  |

### 16.3.1.49 Port B Active Channel Source Address Register (PBACSAADDR)

**Figure 16-65. Port B Active Channel Source Address Register (PBACSAADDR) [offset = 198h]**



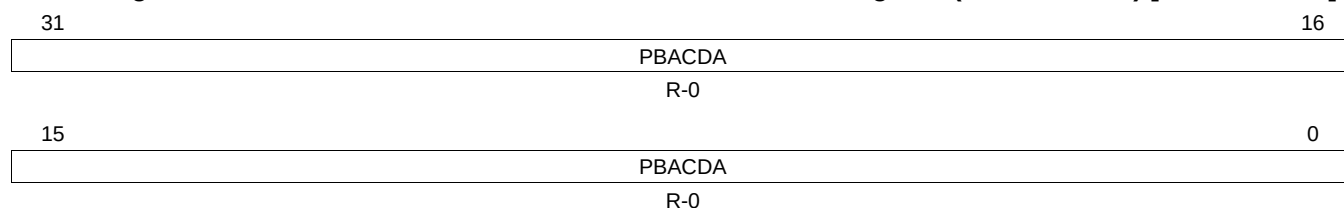
LEGEND: R = Read only; -n = value after reset

**Table 16-57. Port B Active Channel Source Address Register (PBACSAADDR) Field Descriptions**

| Bit  | Field  | Value        | Description                                                                                                                                                                  |
|------|--------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | PBACSA | 0-FFFF FFFFh | Port B Active Channel Source Address. This register contains the current source address of the active channel as broadcasted in <a href="#">Section 16.3.1.3</a> for Port B. |

### 16.3.1.50 Port B Active Channel Destination Address Register (PBACDAADDR)

**Figure 16-66. Port B Active Channel Destination Address Register (PBACDAADDR) [offset = 19Ch]**



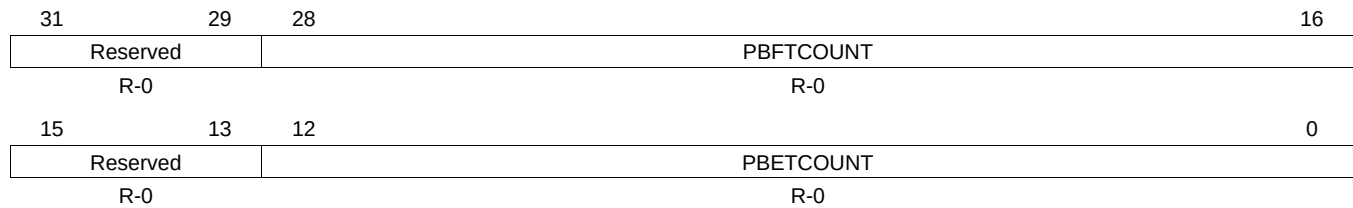
LEGEND: R = Read only; -n = value after reset

**Table 16-58. Port B Active Channel Destination Address Register (PBACDAADDR) Field Descriptions**

| Bit  | Field  | Value        | Description                                                                                                                                                                            |
|------|--------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | PBACDA | 0-FFFF FFFFh | Port B Active Channel Destination Address. This register contains the current destination address of the active channel as broadcasted in <a href="#">Section 16.3.1.3</a> for Port B. |

### 16.3.1.51 Port B Active Channel Transfer Count Register (PBACTC)

**Figure 16-67. Port B Active Channel Transfer Count Register (PBACTC) [offset = 1A0h]**



LEGEND: R = Read only; -n = value after reset

**Table 16-59. Port B Active Channel Transfer Count Register (PBACTC) Field Descriptions**

| Bit   | Field     | Value   | Description                                                                                                                                                                  |
|-------|-----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-29 | Reserved  | 0       | Reads return 0. Writes have no effect.                                                                                                                                       |
| 28-16 | PBFTCOUNT | 0-1FFFh | Port B active channel frame count. These bits contain the current frame count value of the active channel as broadcasted in <a href="#">Section 16.3.1.3</a> for Port B.     |
| 15-13 | Reserved  | 0       | Reads return 0. Writes have no effect.                                                                                                                                       |
| 12-0  | PBETCOUNT | 0-1FFFh | Port B active channel element count. These bits contain the current element count value of the active channel as broadcasted in <a href="#">Section 16.3.1.3</a> for Port B. |

### 16.3.1.52 Parity Control Register (DMAPCR)

**Figure 16-68. Parity Control Register (DMAPCR) [offset = 1A8h]**

|          |  |  |  |  |        |   |          |  |  |  |            |    |        |
|----------|--|--|--|--|--------|---|----------|--|--|--|------------|----|--------|
| 31       |  |  |  |  |        |   |          |  |  |  |            | 15 | 16     |
| Reserved |  |  |  |  |        |   |          |  |  |  |            |    | ERRA   |
| R-0      |  |  |  |  |        |   |          |  |  |  |            |    | R/WP-0 |
| 15       |  |  |  |  | 9      | 8 | 7        |  |  |  | 4          | 3  | 0      |
| Reserved |  |  |  |  | TEST   |   | Reserved |  |  |  | PARITY_ENA |    |        |
| R-0      |  |  |  |  | R/WP-0 |   | R-0      |  |  |  | R/WP-5h    |    |        |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-60. Parity Control Register (DMAPCR) Field Descriptions**

| Bit   | Field      | Value            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-17 | Reserved   | 0                | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 16    | ERRA       | 0                | Error action.                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|       |            | 1                | If a parity error is detected on control packet x (x = 0, 1, ... n), then the enable/disable state of control packet x remains unchanged.                                                                                                                                                                                                                                                                                                                    |
|       |            |                  | If a parity error is detected on control packet x (x = 0, 1, ...n), then the DMA controller is disabled immediately. If a frame on control packet x is processed at the time the parity error is detected, then remaining elements of this frame will not be transferred anymore. The DMA will be disabled regardless of whether the error was detected during a read to the control packet RAM performed by the DMA state machine or by a different master. |
| 15-9  | Reserved   | 0                | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8     | TEST       | 0                | When this bit is set, the parity bits are memory mapped to make them accessible by the CPU.                                                                                                                                                                                                                                                                                                                                                                  |
|       |            | 1                | The parity bits are not memory mapped.                                                                                                                                                                                                                                                                                                                                                                                                                       |
|       |            |                  | The parity bits are memory mapped.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 7-4   | Reserved   | 0                | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3-0   | PARITY_ENA | 5h               | Parity error detection enable. This bit field enables or disables the parity check on read operations and the parity calculation on write operations. If parity checking is enabled and a parity error is detected, the DMA_UERR signal is activated.                                                                                                                                                                                                        |
|       |            | All other values | The parity check is disabled.                                                                                                                                                                                                                                                                                                                                                                                                                                |
|       |            |                  | The parity check is enabled.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|       |            |                  | <b>Note: It is recommended to write Ah to enable parity check, to guard against soft error from flipping PARITY_ENA to a disable state.</b>                                                                                                                                                                                                                                                                                                                  |

### 16.3.1.53 DMA Parity Error Address Register (DMAPAR)

**Figure 16-69. DMA Parity Error Address Register (DMAPAR) [offset = 1ACh]**

|          |    |              |          |    |
|----------|----|--------------|----------|----|
| 31       | 25 | 24           | 23       | 16 |
| Reserved |    | EDFLAG       | Reserved |    |
| R-0      |    | R/W1C-0      | R-0      |    |
| 15       | 12 | 11           | 0        |    |
| Reserved |    | ERRORADDRESS |          |    |
| R-0      |    | R-X          |          |    |

LEGEND: R/W = Read/Write; R = Read only; W1C = Write 1 to clear; X= Undefined; -n = value after reset

**Table 16-61. DMA Parity Error Address Register (DMAPAR) Field Descriptions**

| Bit   | Field        | Value  | Description                                                                                                                                                                                                                                                                                                                                                      |
|-------|--------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-25 | Reserved     | 0      | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                           |
| 24    | EDFLAG       | 0      | Parity Error Detection Flag. This flag indicates if a parity error occurred on reading DMA Control packet RAM.<br>Read: No error occurred.<br>Write: No effect.                                                                                                                                                                                                  |
|       |              | 1      | Read: Error detected and the address is captured in DMAPAR's ERROR_ADDRESS field<br>Write: Clears the bit.                                                                                                                                                                                                                                                       |
| 23-12 | Reserved     | 0      | Reads return 0. Writes have no effect.                                                                                                                                                                                                                                                                                                                           |
| 11-0  | ERRORADDRESS | 0-FFFh | Error address. These bits hold the address of the first parity error generated in the RAM. This error address is frozen from being updated until it is read by the CPU. During emulation mode when SUSPEND is high, this address is frozen even when read.<br><b>Note: The error address register will not be reset by PORRST nor by any other reset source.</b> |

### 16.3.1.54 DMA Memory Protection Control Register (DMAMPCTRL)

**Figure 16-70. DMA Memory Protection Control Register (DMAMPCTRL) [offset = 1B0h]**

|          |        |         |         |         |    |    |
|----------|--------|---------|---------|---------|----|----|
| 31       | 29     | 28      | 27      | 26      | 25 | 24 |
| Reserved | INT3AB | INT3ENA | REG3AP  | REG3ENA |    |    |
| R-0      | R/WP-0 | R/WP-0  | R/WP-0  | R/WP-0  |    |    |
| 23       | 21     | 20      | 19      | 18      | 17 | 16 |
| Reserved | INT2AB | INT2ENA | REG2AP  | REG2ENA |    |    |
| R-0      | R/WP-0 | R/WP-0  | R/WP-0  | R/WP-0  |    |    |
| 15       | 13     | 12      | 11      | 10      | 9  | 8  |
| Reserved | INT1AB | INT1ENA | REG1AP  | REG1ENA |    |    |
| R-0      | R/WP-0 | R/WP-0  | R/WP-0  | R/WP-0  |    |    |
| 7        | 5      | 4       | 3       | 2       | 1  | 0  |
| Reserved | INT0AB | INT0ENA | REG0AP  | REG0ENA |    |    |
| R-0      | R/WP-0 | R/WP-0  | R/WP-00 | R/WP-0  |    |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-62. DMA Memory Protection Control Register (DMAMPCTRL) Field Descriptions**

| Bit   | Field    | Value               | Description                                                                                                                                                                                                          |
|-------|----------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-29 | Reserved | 0                   | Reads return 0. Writes have no effect.                                                                                                                                                                               |
| 28    | INT3AB   | 0<br>1              | Interrupt assignment of region 3 to Group A or Group B.<br>The interrupt is routed to the VIM (Group A).<br>The interrupt is routed to the second CPU (Group B).                                                     |
| 27    | INT3ENA  | 0<br>1              | Interrupt enable of region 3.<br>The interrupt is disabled.<br>The interrupt is enabled.                                                                                                                             |
| 26-25 | REG3AP   | 0<br>1h<br>2h<br>3h | Region 3 access permission. These bits determine the access permission for region 3.<br>All accesses are allowed.<br>Read only accesses are allowed.<br>Write only accesses are allowed.<br>No accesses are allowed. |
| 24    | REG3ENA  | 0<br>1              | Region 3 enable.<br>The region is disabled (no address checking done).<br>The region is enabled (address and access permission checking done).                                                                       |
| 23-21 | Reserved | 0                   | Reads return 0. Writes have no effect.                                                                                                                                                                               |
| 20    | INT2AB   | 0<br>1              | Interrupt assignment of region 2 to Group A or Group B.<br>The interrupt is routed to the VIM (Group A).<br>The interrupt is routed to the second CPU (Group B).                                                     |
| 19    | INT2ENA  | 0<br>1              | Interrupt enable of region 2.<br>The interrupt is disabled.<br>The interrupt is enabled.                                                                                                                             |
| 18-17 | REG2AP   | 0<br>1h<br>2h<br>3h | Region 2 access permission. These bits determine the access permission for region 2.<br>All accesses are allowed.<br>Read only accesses are allowed.<br>Write only accesses are allowed.<br>No accesses are allowed. |
| 16    | REG2ENA  | 0<br>1              | Region 2 enable.<br>The region is disabled (no address checking done).<br>The region is enabled (address and access permission checking done).                                                                       |

**Table 16-62. DMA Memory Protection Control Register (DMAMPCTRL) Field Descriptions (continued)**

| Bit   | Field    | Value               | Description                                                                                                                                                                                                          |
|-------|----------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-13 | Reserved | 0                   | Reads return 0. Writes have no effect.                                                                                                                                                                               |
| 12    | INT1AB   | 0<br>1              | Interrupt assignment of region 1 to Group A or Group B.<br>The interrupt is routed to the VIM (Group A).<br>The interrupt is routed to the second CPU (Group B).                                                     |
| 11    | INT1ENA  | 0<br>1              | Interrupt enable of region 1.<br>The interrupt is disabled.<br>The interrupt is enabled.                                                                                                                             |
| 10-9  | REG1AP   | 0<br>1h<br>2h<br>3h | Region 1 access permission. These bits determine the access permission for region 1.<br>All accesses are allowed.<br>Read only accesses are allowed.<br>Write only accesses are allowed.<br>No accesses are allowed. |
| 8     | REG1ENA  | 0<br>1              | Region 1 enable.<br>The region is disabled (no address checking done).<br>The region is enabled (address and access permission checking done).                                                                       |
| 7-5   | Reserved | 0                   | Reads return 0. Writes have no effect.                                                                                                                                                                               |
| 4     | INT0AB   | 0<br>1              | Interrupt assignment of region 0 to Group A or Group B.<br>The interrupt is routed to the VIM (Group A).<br>The interrupt is routed to the DSP CPU (Group B).                                                        |
| 3     | INT0ENA  | 0<br>1              | Interrupt enable of region 0.<br>The interrupt is disabled.<br>The interrupt is enabled.                                                                                                                             |
| 2-1   | REG0AP   | 0<br>1h<br>2h<br>3h | Region 0 access permission. These bits determine the access permission for region 0.<br>All accesses are allowed.<br>Read only accesses are allowed.<br>Write only accesses are allowed.<br>No accesses are allowed. |
| 0     | REG0ENA  | 0<br>1              | Region 0 enable.<br>The region is disabled (no address checking done).<br>The region is enabled (address and access permission checking done).                                                                       |

### 16.3.1.55 DMA Memory Protection Status Register (DMAMPST)

**Figure 16-71. DMA Memory Protection Status Register (DMAMPST) [offset = 1B4h]**

|          |         |          |         |    |    |
|----------|---------|----------|---------|----|----|
| 31       | 25      | 24       | 23      | 17 | 16 |
| Reserved | REG3FT  | Reserved | REG2FT  |    |    |
| R-0      | R/W1C-0 | R-0      | R/W1C-0 |    |    |
| 15       | 9       | 8        | 7       | 1  | 0  |
| Reserved | REG1FT  | Reserved | REG0FT  |    |    |
| R-0      | R/W1C-0 | R-0      | R/W1C-0 |    |    |

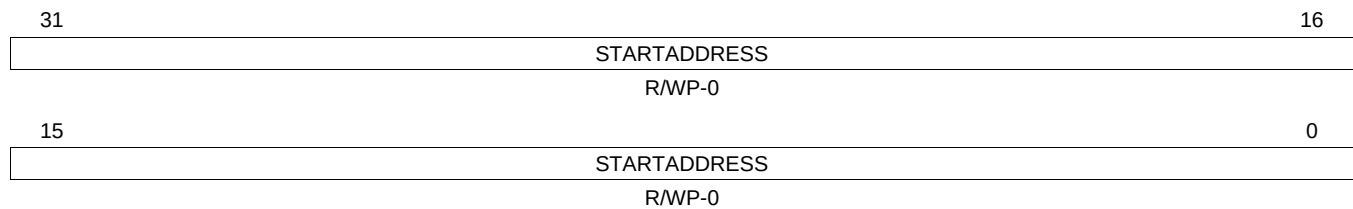
LEGEND: R/W = Read/Write; R = Read only; W1C = Write 1 to clear; -n = value after reset

**Table 16-63. DMA Memory Protection Status Register (DMAMPST) Field Descriptions**

| Bit   | Field    | Value | Description                                                                                                                                                  |
|-------|----------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-25 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                       |
| 24    | REG3FT   | 0     | Region 3 fault. This bit determines whether an access permission violation was detected in this region.<br>Read: No fault was detected.<br>Write: No effect. |
|       |          | 1     | Read: A fault was detected.<br>Write: The bit was cleared.                                                                                                   |
| 23-17 | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                       |
| 16    | REG2FT   | 0     | Region 2 fault. This bit determines whether an access permission violation was detected in this region.<br>Read: No fault was detected.<br>Write: No effect. |
|       |          | 1     | Read: A fault was detected.<br>Write: The bit was cleared.                                                                                                   |
| 15-9  | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                       |
| 8     | REG1FT   | 0     | Region 1 fault. This bit determines whether an access permission violation was detected in this region.<br>Read: No fault was detected.<br>Write: No effect. |
|       |          | 1     | Read: A fault was detected.<br>Write: The bit was cleared.                                                                                                   |
| 7-1   | Reserved | 0     | Reads return 0. Writes have no effect.                                                                                                                       |
| 0     | REG0FT   | 0     | Region 0 fault. This bit determines whether an access permission violation was detected in this region.<br>Read: No fault was detected.<br>Write: No effect. |
|       |          | 1     | Read: A fault was detected.<br>Write: The bit was cleared.                                                                                                   |

### 16.3.1.56 DMA Memory Protection Region 0 Start Address Register (DMAMPR0S)

**Figure 16-72. DMA Memory Protection Region 0 Start Address Register (DMAMPR0S)**  
[offset = 1B8h]



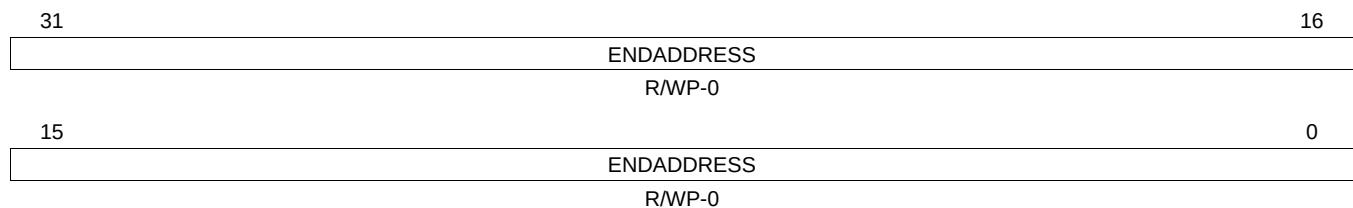
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-64. DMA Memory Protection Region 0 Start Address Register (DMAMPR0S)**  
**Field Descriptions**

| Bit  | Field        | Value        | Description                                                                                                                                                 |
|------|--------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | STARTADDRESS | 0-FFFF FFFFh | Start Address defines the address at which the region begins. The effective start address is truncated to the nearest word address, that is, 0x103 = 0x100. |

### 16.3.1.57 DMA Memory Protection Region 0 End Address Register (DMAMPR0E)

**Figure 16-73. DMA Memory Protection Region 0 End Address Register (DMAMPR0E)**  
[offset = 1BCh]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-65. DMA Memory Protection Region 0 End Address Register (DMAMPR0E)**  
**Field Descriptions**

| Bit  | Field      | Value        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | ENDADDRESS | 0-FFFF FFFFh | <p>End Address defines the address at which the region ends. The end address usually is larger than the start address for this region; otherwise, the region will wrap around at the end of the address space. The end address is the start address plus the region length minus 1. The effective end address is rounded up to the nearest 32-bit word end address, that is, 0x200 = 0x203.</p> <p><b>Note:</b> When using 64-bit transfers, the address is rounded up to the nearest 64-bit word end address, that is, 0x200 = 0x207. All other transfers are rounded up to the nearest 32-bit word end address.</p> |

### 16.3.1.58 DMA Memory Protection Region 1 Start Address Register (DMAMPR1S)

**Figure 16-74. DMA Memory Protection Region 1 Start Address Register (DMAMPR1S)**  
[offset = 1C0h]

|              |    |
|--------------|----|
| 31           | 16 |
| STARTADDRESS |    |
| R/WP-0       |    |
| 15           | 0  |
| STARTADDRESS |    |
| R/WP-0       |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-66. DMA Memory Protection Region 1 Start Address Register (DMAMPR1S)**  
**Field Descriptions**

| Bit  | Field        | Value        | Description                                                                                                                                                 |
|------|--------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | STARTADDRESS | 0-FFFF FFFFh | Start Address defines the address at which the region begins. The effective start address is truncated to the nearest word address, that is, 0x103 = 0x100. |

### 16.3.1.59 DMA Memory Protection Region 1 End Address Register (DMAMPR1E)

**Figure 16-75. DMA Memory Protection Region 1 End Address Register (DMAMPR1E)**  
[offset = 1C4h]

|            |    |
|------------|----|
| 31         | 16 |
| ENDADDRESS |    |
| R/WP-0     |    |
| 15         | 0  |
| ENDADDRESS |    |
| R/WP-0     |    |

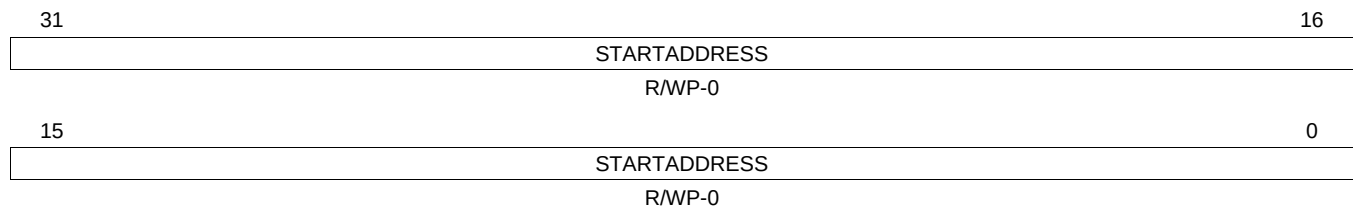
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-67. DMA Memory Protection Region 1 End Address Register (DMAMPR1E)**  
**Field Descriptions**

| Bit  | Field      | Value        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | ENDADDRESS | 0-FFFF FFFFh | End Address defines the address at which the region ends. The end address usually is larger than the start address for this region; otherwise, the region will wrap around at the end of the address space. The end address is the start address plus the region length minus 1. The effective end address is rounded up to the nearest 32-bit word end address, that is, 0x200 = 0x203.<br><br><b>Note: When using 64-bit transfers, the address is rounded up to the nearest 64-bit word end address, that is, 0x200 = 0x207. All other transfers are rounded up to the nearest 32-bit word end address.</b> |

### 16.3.1.60 DMA Memory Protection Region 2 Start Address Register (DMAMPR2S)

**Figure 16-76. DMA Memory Protection Region 2 Start Address Register (DMAMPR2S)**  
[offset = 1C8h]



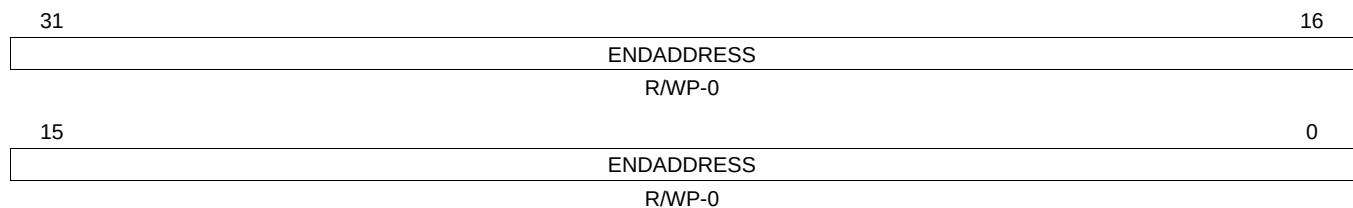
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-68. DMA Memory Protection Region 2 Start Address Register (DMAMPR2S)**  
**Field Descriptions**

| Bit  | Field        | Value        | Description                                                                                                                                                 |
|------|--------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | STARTADDRESS | 0-FFFF FFFFh | Start Address defines the address at which the region begins. The effective start address is truncated to the nearest word address, that is, 0x103 = 0x100. |

### 16.3.1.61 DMA Memory Protection Region 2 End Address Register (DMAMPR2E)

**Figure 16-77. DMA Memory Protection Region 2 End Address Register (DMAMPR2E)**  
[offset = 1CCh]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-69. DMA Memory Protection Region 2 End Address Register (DMAMPR2E)**  
**Field Descriptions**

| Bit  | Field      | Value        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | ENDADDRESS | 0-FFFF FFFFh | <p>End Address defines the address at which the region ends. The end address usually is larger than the start address for this region; otherwise, the region will wrap around at the end of the address space. The end address is the start address plus the region length minus 1. The effective end address is rounded up to the nearest 32-bit word end address, that is, 0x200 = 0x203.</p> <p><b>Note:</b> When using 64-bit transfers, the address is rounded up to the nearest 64-bit word end address, that is, 0x200 = 0x207. All other transfers are rounded up to the nearest 32-bit word end address.</p> |

### 16.3.1.62 DMA Memory Protection Region 3 Start Address Register (DMAMPR3S)

**Figure 16-78. DMA Memory Protection Region 3 Start Address Register (DMAMPR3S)**  
[offset = 1D0h]

|              |    |
|--------------|----|
| 31           | 16 |
| STARTADDRESS |    |
| R/WP-0       |    |
| 15           | 0  |
| STARTADDRESS |    |
| R/WP-0       |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-70. DMA Memory Protection Region 3 Start Address Register (DMAMPR3S)**  
**Field Descriptions**

| Bit  | Field        | Value        | Description                                                                                                                                                 |
|------|--------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | STARTADDRESS | 0-FFFF FFFFh | Start Address defines the address at which the region begins. The effective start address is truncated to the nearest word address, that is, 0x103 = 0x100. |

### 16.3.1.63 DMA Memory Protection Region 3 End Address Register (DMAMPR3E)

**Figure 16-79. DMA Memory Protection Region 3 End Address Register (DMAMPR3E)**  
[offset = 1D4h]

|            |    |
|------------|----|
| 31         | 16 |
| ENDADDRESS |    |
| R/WP-0     |    |
| 15         | 0  |
| ENDADDRESS |    |
| R/WP-0     |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

**Table 16-71. DMA Memory Protection Region 3 End Address Register (DMAMPR3E)**  
**Field Descriptions**

| Bit  | Field      | Value        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | ENDADDRESS | 0-FFFF FFFFh | End Address defines the address at which the region ends. The end address usually is larger than the start address for this region; otherwise, the region will wrap around at the end of the address space. The end address is the start address plus the region length minus 1. The effective end address is rounded up to the nearest 32-bit word end address, that is, 0x200 = 0x203.<br><br><b>Note: When using 64-bit transfers, the address is rounded up to the nearest 64-bit word end address, that is, 0x200 = 0x207. All other transfers are rounded up to the nearest 32-bit word end address.</b> |

### 16.3.2 Channel Configuration

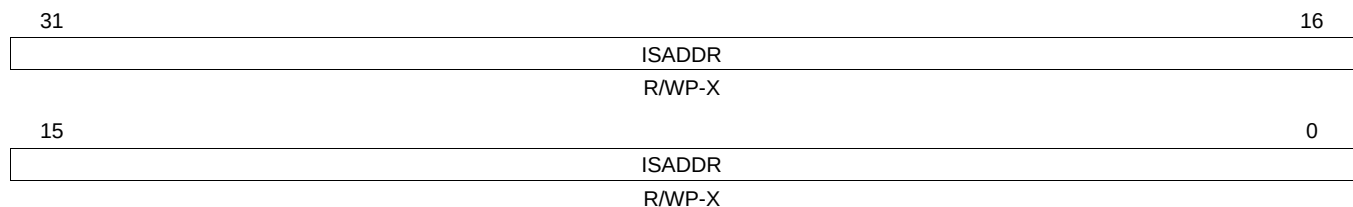
The channel configuration is defined by the channel control packet: channel control, transfer count, offset values, source/destination address.

- It is stored in local RAM, which is protected by parity.
- Each control packet contains a total of nine fields.
- The first six fields are programmable, while the last three fields are read only.
- The RAM is accessible by queue A and queue B state machines as well as CPU.
- When there are simultaneous accesses, the priority is resolved in a fixed priority scheme with the CPU having the highest priority.

All the control packets look the same. Following, there is the detailed layout of these registers shown for control packet 0.

#### 16.3.2.1 Initial Source Address (ISADDR)

**Figure 16-80. Initial Source Address (ISADDR) [offset = 00]**



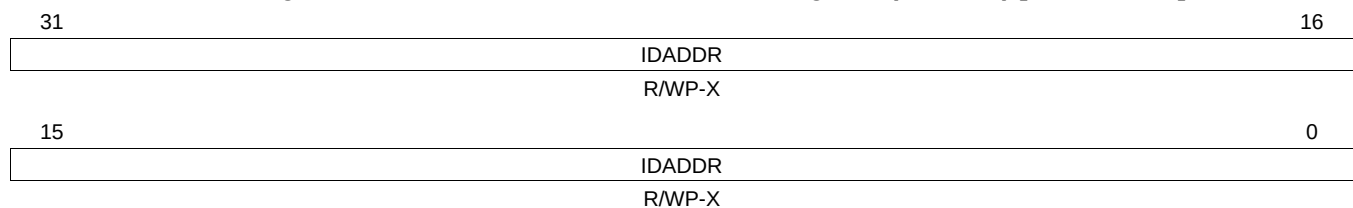
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset; X = Unknown

**Table 16-72. Initial Source Address (ISADDR) Field Descriptions**

| Bit  | Field  | Value        | Description                                                                            |
|------|--------|--------------|----------------------------------------------------------------------------------------|
| 31-0 | ISADDR | 0-FFFF FFFFh | Initial source address. These bits give the absolute 32-bit source address (physical). |

#### 16.3.2.2 Initial Destination Address Register (IDADDR)

**Figure 16-81. Initial Destination Address Register (IDADDR) [offset = 04h]**



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset; X = Unknown

**Table 16-73. Initial Destination Address Register (IDADDR) Field Descriptions**

| Bit  | Field  | Value        | Description                                                                                      |
|------|--------|--------------|--------------------------------------------------------------------------------------------------|
| 31-0 | IDADDR | 0-FFFF FFFFh | Initial destination address. These bits give the absolute 32-bit destination address (physical). |

### 16.3.2.3 Initial Transfer Count Register (ITCOUNT)

**Figure 16-82. Initial Transfer Count Register (ITCOUNT) [offset = 08h]**

|          |    |          |    |
|----------|----|----------|----|
| 31       | 29 | 28       | 16 |
| Reserved |    | IFTCOUNT |    |
| R-X      |    | R/WP-X   |    |
| 15       | 13 | 12       | 0  |
| Reserved |    | IETCOUNT |    |
| R-X      |    | R/WP-X   |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset; X = Unknown

**Table 16-74. Initial Transfer Count Register (ITCOUNT) Field Descriptions**

| Bit   | Field    | Value   | Description                                                                                                                            |
|-------|----------|---------|----------------------------------------------------------------------------------------------------------------------------------------|
| 31-29 | Reserved | 0       | Reads are undefined. Writes have no effect.                                                                                            |
| 28-16 | IFTCOUNT | 0-1FFFh | Initial frame transfer count. These bits define the number of frame transfers.                                                         |
| 15-13 | Reserved | 0       | Reads are undefined. Writes have no effect.                                                                                            |
| 12-0  | IETCOUNT | 0-1FFFh | Initial element transfer count. These bits define the number of element transfers. The block transfer size will be IETCOUNT x IFTCOUNT |

### 16.3.2.4 Channel Control Register (CHCTRL)

**Figure 16-83. Channel Control Register (CHCTRL) [offset = 10h]**

|          |          |          |        |
|----------|----------|----------|--------|
| 31       | 22       | 21       | 16     |
| Reserved |          |          |        |
| R-X      |          |          |        |
| CHAIN    |          |          |        |
| R/WP-X   |          |          |        |
| 15       | 14       | 13       | 12     |
| 11       | 9        | 8        | 7      |
| 5        | 4        | 3        | 2      |
| 1        | 0        |          |        |
| RES      | WES      | Reserved | TTYPE  |
| Reserved | Reserved | ADDMR    | ADDMW  |
| AIM      |          |          |        |
| R/WP-X   | R/WP-X   | R-X      | R/WP-X |
| R/WP-X   | R/WP-X   | R/WP-X   | R/WP-X |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset; X = Unknown

**Table 16-75. Channel Control Register (CHCTRL) Field Descriptions**

| Bit   | Field    | Value                          | Description                                                                                                                                                                                                                                                                                                                                                                |
|-------|----------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-22 | Reserved | 0                              | Reads are undefined. Writes have no effect.                                                                                                                                                                                                                                                                                                                                |
| 21-16 | CHAIN    | 0<br>1h<br>:<br>10h<br>11h-3Fh | Next channel to be triggered. At the end of the programmed number of frames, the specified channel will be triggered. Programmed value x means channel (x-1) is chained.<br><br><b>Note: The programmer must program the CHAIN bits before initiating a DMA transfer.</b><br>No channel is selected.<br>Channel 0 is selected.<br>:<br>Channel 15 is selected.<br>Reserved |
| 15-14 | RES      | 0<br>1h<br>2h<br>3h            | Read element size.<br>The element is byte, 8-bit.<br>The element is half-word, 16-bit.<br>The element is word, 32-bit.<br>The element is double-word, 64-bit.                                                                                                                                                                                                              |
| 13-12 | WES      | 0<br>1h<br>2h<br>3h            | Write element size.<br>The element is byte, 8-bit.<br>The element is half-word, 16-bit.<br>The element is word, 32-bit.<br>The element is double-word, 64-bit.                                                                                                                                                                                                             |
| 11-9  | Reserved | 0                              | Reads are undefined. Writes have no effect.                                                                                                                                                                                                                                                                                                                                |
| 8     | TTYPE    | 0<br>1                         | Transfer type.<br>A request triggers one frame transfer.<br>A request triggers one block transfer.                                                                                                                                                                                                                                                                         |
| 7-5   | Reserved | 0                              | Reads are undefined. Writes have no effect.                                                                                                                                                                                                                                                                                                                                |
| 4-3   | ADDMR    | 0<br>1h<br>2h<br>3h            | Addressing mode read.<br>Constant<br>Post-increment<br>Reserved<br>Indexed                                                                                                                                                                                                                                                                                                 |
| 2-1   | ADDMW    | 0<br>1h<br>2h<br>3h            | Addressing mode write.<br>Constant<br>Post-increment<br>Reserved<br>Indexed                                                                                                                                                                                                                                                                                                |
| 0     | AIM      | 0<br>1                         | Auto-initiation mode.<br>Auto-initiation mode is disabled.<br>Auto-initiation mode is enabled.                                                                                                                                                                                                                                                                             |

### 16.3.2.5 Element Index Offset Register (EIOFF)

**Figure 16-84. Element Index Offset Register (EIOFF) [offset = 14h]**

|          |        |    |    |
|----------|--------|----|----|
| 31       | 29     | 28 | 16 |
| Reserved | EIDX   |    |    |
| R-X      | R/WP-X |    |    |
| 15       | 13     | 12 | 0  |
| Reserved | EIDX   |    |    |
| R-X      | R/WP-X |    |    |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset; X = Unknown

**Table 16-76. Element Index Offset Register (EIOFF) Field Descriptions**

| Bit   | Field    | Value   | Description                                                                                                                         |
|-------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------|
| 31-29 | Reserved | 0       | Reads are undefined. Writes have no effect.                                                                                         |
| 28-16 | EIDX     | 0-1FFFh | Destination address element index. These bits define the offset to be added to the destination address after each element transfer. |
| 15-13 | Reserved | 0       | Reads are undefined. Writes have no effect.                                                                                         |
| 12-0  | EIDX     | 0-1FFFh | Source address element index. These bits define the offset to be added to the source address after each element transfer.           |

### 16.3.2.6 Frame Index Offset Register (FIOFF)

**Figure 16-85. Frame Index Offset Register (FIOFF) [offset = 18h]**

|          |        |    |    |
|----------|--------|----|----|
| 31       | 29     | 28 | 16 |
| Reserved | FIDX   |    |    |
| R-X      | R/WP-X |    |    |
| 15       | 13     | 12 | 0  |
| Reserved | FIDX   |    |    |
| R-X      | R/WP-X |    |    |

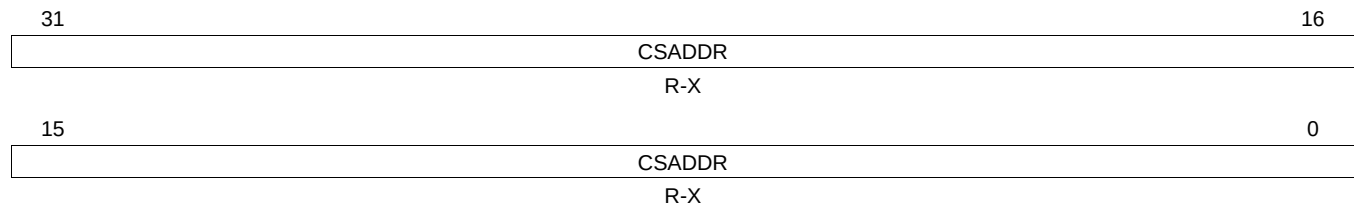
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset; X = Unknown

**Table 16-77. Frame Index Offset Register (FIOFF) Field Descriptions**

| Bit   | Field    | Value   | Description                                                                                                                         |
|-------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------|
| 31-29 | Reserved | 0       | Reads are undefined. Writes have no effect.                                                                                         |
| 28-16 | FIDX     | 0-1FFFh | Destination address frame index. These bits define the offset to be added to the destination address after element count reached 1. |
| 15-13 | Reserved | 0       | Reads are undefined. Writes have no effect.                                                                                         |
| 12-0  | FIDX     | 0-1FFFh | Source address frame index. These bits define the offset to be added to the source address after element count reached 1.           |

### 16.3.2.7 Current Source Address Register (CSADDR)

**Figure 16-86. Current Source Address Register (CSADDR) [offset = 800h]**



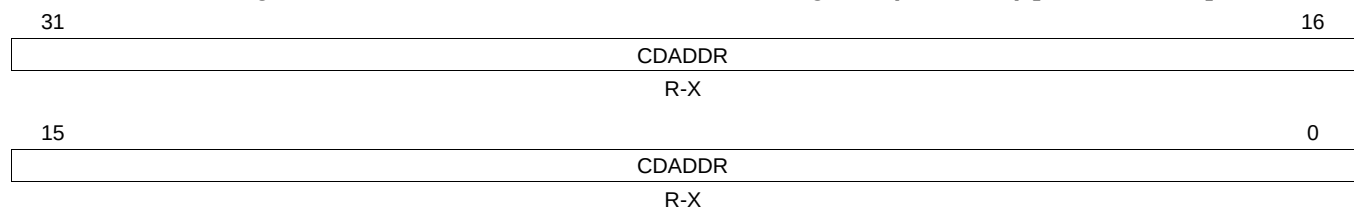
LEGEND: R = Read only; -n = value after reset; X = Unknown

**Table 16-78. Current Source Address Register (CSADDR) Field Descriptions**

| Bit  | Field  | Value        | Description                                                                                                                                                                                      |
|------|--------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | CSADDR | 0-FFFF FFFFh | Current source address. These bits contain the current working absolute 32-bit source address (physical). These bits are only updated after a channel is arbitrated out from the priority queue. |

### 16.3.2.8 Current Destination Address Register (CDADDR)

**Figure 16-87. Current Destination Address Register (CDADDR) [offset = 804h]**



LEGEND: R = Read only; -n = value after reset; X = Unknown

**Table 16-79. Current Destination Address Register (CDADDR) Field Descriptions**

| Bit  | Field  | Value        | Description                                                                                                                                                                                              |
|------|--------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-0 | CDADDR | 0-FFFF FFFFh | Current destination address. These bits contain the current working absolute 32-bit destination address (physical). These bits are only updated after a channel is arbitrated out of the priority queue. |

### 16.3.2.9 Current Transfer Count Register (CTCOUNT)

**Figure 16-88. Current Transfer Count Register (CTCOUNT) [offset = 808h]**

|          |    |          |    |
|----------|----|----------|----|
| 31       | 29 | 28       | 16 |
| Reserved |    | CFTCOUNT |    |
| R-X      |    | R-X      |    |
| 15       | 13 | 12       | 0  |
| Reserved |    | CETCOUNT |    |
| R-X      |    | R-X      |    |

LEGEND: R = Read only; -n = value after reset; X = Unknown

**Table 16-80. Current Transfer Count Register (CTCOUNT) Field Descriptions**

| Bit   | Field    | Value   | Description                                                                                                                                                                       |
|-------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-29 | Reserved | 0       | Reads are undefined. Writes have no effect.                                                                                                                                       |
| 28-16 | CFTCOUNT | 0-1FFFh | Current frame transfer count. Returned the current remaining frame counts.                                                                                                        |
| 15-13 | Reserved | 0       | Reads are undefined. Writes have no effect.                                                                                                                                       |
| 12-0  | CETCOUNT | 0-1FFFh | Current element transfer count. These bits return the current remaining element counts. CTCOUNT register is only updated after a channel is arbitrated out of the priority queue. |